

Endogenous Switching Regression with Heterogeneous Selection on Gains: Assumptions, Tests, and Estimation

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October 2026

Software: `esreg` 1.0.0 Zenodo: <https://doi.org/10.5281/zenodo.22717028>

Abstract

The endogenous switching regression is the one workhorse model in which selection on gains—the covariance κ between the gain from treatment and the unobservable that drives participation—is a free parameter rather than an assumption. This paper asks under what conditions the selection equation identifies κ , how each condition can be checked, and which estimation route to take when a condition fails. It orders the assumptions by what they cost: a normal single-index selection equation, a linear conditional mean of the regime errors given the participation error, and trivariate normality, together with the exclusion, sufficiency and variation conditions on the instruments. It then gives one test per assumption, all borne by the selection equation: a link test and a probit-versus-FIML contrast for the index, a test of index sufficiency for the instruments, diagnostics of strength and support, a Wald test of constant κ in the covariates, a Hausman contrast with an influence-function covariance and Hermite controls that separate a failure of trivariate normality from a failure of the linear conditional mean, and a pseudo-difference-in-differences on the index whose within-group contrasts read selection on gains off a two-by-two table with almost no model. Estimation proceeds by tiers, removing one assumption at a time: full-information maximum likelihood with heterogeneous variances and correlations, a two-step estimator with the exact variance of the whole procedure from stacked moments, the same two-step augmented by Hermite terms when the conditional mean is not linear, and a semiparametric marginal treatment effect on the score. The model extends to $\kappa(x)$, a selection on gains that varies with the covariates. Monte Carlo experiments give the tests their size and show that each rejects what it is built to reject; the manual example of Stata’s `etregress` shows the reading changing the conclusion. The Stata command `esreg` and its family of post-estimation commands implement the whole, with survey weights, clusters and the survey design; they are validated value for value against a Python reference implementation and against `etregress`, `movestay` and `msat`, and their standard errors against brute-force influence functions and a Monte Carlo.

Keywords: endogenous switching regression, selection on gains, marginal treatment effect, specification tests, Stata.

JEL: C21, C31, C52, C87.

1 Introduction

A program that people join voluntarily is evaluated on the outcomes of those who joined and of those who did not, and the difference between the two groups mixes the effect of the program with the reasons for joining. When those reasons are related to what the program would do for each person—when the ones who join are the ones who gain most, or least—the average effect on the treated, the average effect on the untreated and the average effect in the population are three different numbers, and a policy that extends the program to non-participants yields the

second, not the first. [Araar \(2026a\)](#) called this selection on gains, defined it as the covariance κ between the gain from treatment and the unobservable that drives participation, and showed that among the models practitioners use, the endogenous switching regression (ESR) of [Roy \(1951\)](#), [Heckman \(1978\)](#) and [Maddala \(1983\)](#) is the only one in which κ is a parameter to estimate rather than a restriction to impose: matching and inverse-probability weighting set it to zero conditionally on covariates, difference-in-differences differences it away under a common trend, the single-equation treatment regression forces one correlation on both regimes. The price of that freedom is a set of assumptions, and the ESR has a reputation for fragility that comes from them: it is said to rest on trivariate normality and on exclusion restrictions that the data cannot test, and to be identified by functional form when the instrument is weak.

This paper takes the reputation apart. Its starting point is the observation that the fragilities of the ESR are properties of its selection equation, not of the joint normality of its errors taken as a block. Three things are asked of the selection equation: an argument (the exclusion restriction, and the less discussed requirement that the instruments act on the regime errors only through the participation index, which the paper calls index sufficiency), a form (a normal single index, whose Mills ratios are the control functions of both estimation routes), and variation (enough movement of the index within covariate cells for the control function to be told apart from the covariates). The joint law of the regime errors given the participation error is asked for only by the likelihood, in two pieces that can be separated: a linear conditional mean, which the two-step route also needs, and normality of the regime errors around it, which it does not. Ordering the assumptions this way makes each one testable with the material the selection equation provides, and it makes the choice of an estimation route a consequence of the tests rather than a matter of taste.

The paper makes three contributions. The first is a set of tests, one per assumption, all computed from the probit index and the two regime regressions. For the form of the selection equation, a link test and a contrast between the probit and the likelihood estimates of the selection coefficients, which is what trivariate normality imposes on the selection equation. For index sufficiency, the addition of the excluded instruments but one to the regime regressions beside the Mills ratio, whose coefficients are zero under the null and whose sign reads a direct effect of the instrument on the outcome. For the joint law, a Hausman contrast between the likelihood and the two-step estimates with a covariance built from the influence functions of the two estimators—positive semi-definite by construction, valid under the alternative—and Hermite controls beside the Mills ratio that detect a nonlinear conditional mean; the two tests separate the two failures, which matters because they call for different remedies. For the strength and support of the instrument, diagnostics that say how much of the identification rests on the curvature of the Mills ratio. And a model-light test of selection on gains itself: a pseudo-difference-in-differences on strata of the participation index, whose two within-group contrasts have the signs of $\rho_1\sigma_1$ and $\rho_0\sigma_0$ under the linear conditional mean alone, and whose ratio to the X -adjusted contrasts of the Mills ratios is a consistent estimator of κ that never uses the joint law.

The second contribution is $\kappa(x)$, a selection on gains that varies with the covariates: $\rho_j\sigma_j(x) = W(x)'\theta_j$ in the two-step route, where the Mills ratio interacted with W is the regressor and $\kappa(x) = W(x)'(\theta_1 - \theta_0)$ is identified under the same conditions as the constant model plus a rank condition; and heterogeneous $\ln\sigma_j$ and $\operatorname{atanh}\rho_j$ in the likelihood. A program whose selection on gains is stronger among the educated, or among the poor, is then a testable hypothesis, and the marginal treatment effect becomes a surface in (x, p) . The third contribution is the estimation by tiers and its implementation: the likelihood with analytic scores, the two-step with the exact variance of the whole procedure from the stacked moment conditions of the probit and the two regressions—the variance that `msat` ([Araar, 2026b](#)) obtained by bootstrap and that `etregress` provides only for its constrained model—the same two-step augmented by Hermite terms when the conditional mean bends, the effects with the influence function of the whole

procedure, and the semiparametric marginal treatment effect on the score by percentile-weights regression (Araar, 2026c), the tier that keeps only the single index. The Stata command `esreg` and its post-estimation family (`esrdiag`, `esrtest`, `esrcurve`, `esrmte`, `esrreport`) implement all of it, with factor variables, survey weights, clusters and survey designs; every printed number is checked against a Python reference implementation written first, and every standard error against a brute force.

The paper sits next to three literatures. The marginal treatment effect of Heckman and Vytlacil (2005) is the object the semiparametric tier estimates and the object the parametric line summarizes; the ESR is the index model in which the MTE is a line, and κ is its slope. The discrete-instrument literature of Brinch et al. (2017) and the bounds of Mogstad et al. (2018) address the same identification problem from the other side, asking what a discrete instrument can deliver without a parametric line; the paper’s diagnostics of variation and support say where that question bites, and the manual example of Section 7, with a single indicator instrument, shows the semiparametric tier failing for exactly that reason. And the Stata implementations—`movestay` (Lokshin and Sajaia, 2004), `msat` (Araar, 2026b), and the official `etregress` with the `poutcomes` option, which fits the same likelihood (StataCorp, 2025)—estimate the model but do not test it; the paper’s Section 8 sets the four side by side.

Section 2 presents the model, $\kappa(x)$, and the assumptions ordered by cost. Section 3 states what each route identifies and what happens when index sufficiency fails. Section 4 develops the tests. Section 5 describes the estimation by tiers, the stacked variance, the variance of the effects and the augmented two-step. Section 6 reports the Monte Carlo evidence on the tests in one table, and on the standard errors. Section 7 reads the `etregress` manual example with the toolkit, and Section 8 documents the commands, their treatment of survey designs, and their validation. Section 9 concludes. Proofs are in Appendix A, the Monte Carlo designs in Appendix B.

2 The model

2.1 Two regimes and one selection index

The model is the switching regression of Roy (1951), brought into the econometric literature by Heckman (1976, 1978) and Maddala (1983), in the form estimated by Lokshin and Sajaia (2004) and by Stata’s `etregress` with the `poutcomes` option (StataCorp, 2025), and read in Araar (2026a). Each unit has two potential outcomes and one participation decision,

$$Y_1 = X\beta_1 + \omega_1, \quad Y_0 = X\beta_0 + \omega_0, \quad D = \mathbf{1}\{Z\gamma + u > 0\}, \quad Y = DY_1 + (1 - D)Y_0, \quad (1)$$

with X the regressors of the outcome equations (a constant included), $Z \supseteq X$ the regressors of the selection equation, and (ω_1, ω_0, u) the unobservables, independent of Z , with $\text{Var}(u) = 1$. Write $\sigma_j^2 = \text{Var}(\omega_j)$, $\rho_j = \text{Corr}(\omega_j, u)$, $P(Z) = \Pr(D = 1 \mid Z) = \Phi(Z\gamma)$ when u is standard normal, and

$$\lambda_1(v) = \frac{\phi(v)}{\Phi(v)}, \quad \lambda_0(v) = \frac{\phi(v)}{1 - \Phi(v)}, \quad (2)$$

so that $\mathbb{E}[u \mid u > -v] = \lambda_1(v)$ and $\mathbb{E}[u \mid u < -v] = -\lambda_0(v)$. The individual gain is $\Delta = Y_1 - Y_0 = X(\beta_1 - \beta_0) + \omega_1 - \omega_0$, and the parameter of Araar (2026a) is the covariance between its unobserved part and the participation error,

$$\kappa = \text{Cov}(\omega_1 - \omega_0, u) = \rho_1\sigma_1 - \rho_0\sigma_0. \quad (3)$$

When $\mathbb{E}[\omega_j \mid u] = \rho_j\sigma_j u$ the conditional expectations of the outcomes given participation are $\mathbb{E}[Y \mid X, Z, D = 1] = X\beta_1 + \rho_1\sigma_1\lambda_1(Z\gamma)$ and $\mathbb{E}[Y \mid X, Z, D = 0] = X\beta_0 - \rho_0\sigma_0\lambda_0(Z\gamma)$, the expected effects of the treated and of the untreated are $\mathbb{E}[\Delta \mid X, Z, D = 1] = X(\beta_1 - \beta_0) +$

$\kappa\lambda_1(Z\gamma)$ and $\mathbb{E}[\Delta \mid X, Z, D = 0] = X(\beta_1 - \beta_0) - \kappa\lambda_0(Z\gamma)$, and their averages over the two groups satisfy

$$\text{ATT} - \text{ATU} = \kappa(\bar{\lambda}_1 + \bar{\lambda}_0), \quad (4)$$

with $\bar{\lambda}_1$ the mean of $\lambda_1(Z\gamma)$ among the treated and $\bar{\lambda}_0$ the mean of $\lambda_0(Z\gamma)$ among the untreated. The marginal treatment effect at the p -th percentile of the participation unobservable, $p = 0$ being the most eager, is the line

$$\text{MTE}(p) = \mathbb{E}[X](\beta_1 - \beta_0) - \kappa\Phi^{-1}(p), \quad (5)$$

whose slope is $-\kappa$: selection on gains is the tilt of the marginal effect along the participation margin. Everything in (4) and (5) is a consequence of one number, and the paper is about the conditions under which the data deliver that number.

2.2 Heterogeneous selection on gains: $\kappa(x)$

The parameter of [Araar \(2026a\)](#) is a covariance averaged over the population, $\kappa = \text{Cov}(\omega_1 - \omega_0, u) = \rho_1\sigma_1 - \rho_0\sigma_0$. Nothing in the economics of participation says that this covariance is the same for every household. A farmer with little land and a farmer with much land may both know something about their return to a program, but the strength of that private information, and the extent to which it drives participation, can differ with land, education, distance to the market, or any other observable. The object of this paper is therefore the conditional version

$$\kappa(x) = \text{Cov}(\omega_1 - \omega_0, u \mid X = x) = \rho_1\sigma_1(x) - \rho_0\sigma_0(x), \quad (6)$$

where $\rho_j\sigma_j(x) = \text{Cov}(\omega_j, u \mid X = x)$ is the covariance between the regime- j error and the participation error in the subpopulation with covariates x . Selection on gains is heterogeneous when $\kappa(x)$ is not constant; it may change sign, so that some types select on their gains and others against them, and the average $\kappa = \mathbb{E}[\kappa(x)]$ can be close to zero while $\kappa(x)$ is large everywhere.

Two parameterizations. Let $W = W(X)$ be a p_w -vector of known functions of X with a constant first, $W = (1, w(X))$; the leading case is $W = (1, X_{-1})$, the regressors themselves, and $W = 1$ gives back the constant model. The paper uses two ways of letting the selection covariances depend on x .

(i) *The two-step form.* Under Assumptions A1–A2 below, the conditional means in each regime are

$$\mathbb{E}[Y \mid X, Z, D = 1] = X\beta_1 + \rho_1\sigma_1(x)\lambda_1(Z\gamma), \quad \mathbb{E}[Y \mid X, Z, D = 0] = X\beta_0 - \rho_0\sigma_0(x)\lambda_0(Z\gamma), \quad (7)$$

with $\lambda_1(v) = \phi(v)/\Phi(v)$ and $\lambda_0(v) = \phi(v)/(1 - \Phi(v))$. Specifying $\rho_j\sigma_j(x) = W'\theta_j$ turns (7) into two linear regressions of Y on X and on the interactions $W\lambda_j$, and

$$\kappa(x) = W'(\theta_1 - \theta_0), \quad \frac{\partial \kappa(x)}{\partial w_k} = \theta_{1k} - \theta_{0k}. \quad (8)$$

The slope $\theta_{1k} - \theta_{0k}$ is the change in selection on gains per unit of w_k ; its sign says whether the private information that drives participation is stronger among households with more of w_k . The constant model is the restriction $\theta_{1,-1} = \theta_{0,-1}$.

(ii) *The likelihood form.* Under Assumption A3 the joint law of (ω_1, ω_0, u) given X is normal, and heterogeneity is placed on its parameters,

$$\ln \sigma_j = W'_s \alpha_j, \quad \text{atanh } \rho_j = W'_r \delta_j, \quad j = 0, 1, \quad (9)$$

with W_s and W_r two (possibly different) subvectors of W . Then $\kappa(x) = \tanh(W'_r \delta_1) \exp(W'_s \alpha_1) - \tanh(W'_r \delta_0) \exp(W'_s \alpha_0)$, a smooth but nonlinear function of x , and the constant model is nested

as $\alpha_{j,-1} = \delta_{j,-1} = 0$. The two forms coincide when $W_s = W_r = W$ and $W'\theta_j$ is taken as the first-order expansion of $\tanh(W'_r\delta_j) \exp(W'_s\alpha_j)$; in practice they give close estimates of $\kappa(x)$ when the model is right and diverge when the regime laws are not normal, for the reason given in Section 5.

What changes. The individual expected effect keeps the form of the constant model with $\kappa(x)$ in place of κ ,

$$\begin{aligned}\mathbb{E}[Y_1 - Y_0 \mid X, Z, D = 1] &= X(\beta_1 - \beta_0) + \kappa(x)\lambda_1(Z\gamma), \\ \mathbb{E}[Y_1 - Y_0 \mid X, Z, D = 0] &= X(\beta_1 - \beta_0) - \kappa(x)\lambda_0(Z\gamma),\end{aligned}\tag{10}$$

so that the gap between the treated and the untreated at the same x is $\kappa(x)(\lambda_1 + \lambda_0)$, and the marginal treatment effect becomes a surface,

$$\text{MTE}(x, p) = X(\beta_1 - \beta_0) - \kappa(x) \Phi^{-1}(p),\tag{11}$$

whose slope in the participation percentile p is $-\kappa(x)$. With $\kappa(x) > 0$ the early participants at x gain more than the late ones; with $\kappa(x) < 0$ the order is reversed; where $\kappa(x)$ crosses zero the surface is flat in p and the treated and untreated at that x have the same expected gain. The average parameters of the constant model are the X -averages of (10), and κ reported by `esreg` is $\mathbb{E}[\kappa(x)]$ over the estimation sample, so that the identity $\text{ATT} - \text{ATU} = \mathbb{E}[\kappa(x)(\lambda_1 + \lambda_0)]$ replaces $\kappa(\bar{\lambda}_1 + \bar{\lambda}_0)$.

Identification. The two-step form is identified under the same conditions as the constant model, plus a rank condition on W .

Proposition 1 (Identification of $\kappa(x)$). *Let Assumptions A1 (selection error normal, single index) and A2 (linear conditional means, $\mathbb{E}[\omega_j \mid X, Z, u] = \rho_j \sigma_j(X)u$) hold with $\rho_j \sigma_j(x) = W(x)'\theta_j$, and let E (exclusion) and V (variation) hold: Z contains at least one variable excluded from X whose coefficient in γ is non-zero, and $\text{Var}(Z\gamma \mid X = x, D = j) > 0$ for almost every x , $j = 0, 1$. If $\mathbb{E}[XX' \mid D = j]$ and $\mathbb{E}[WW' \mid D = j]$ are nonsingular for $j = 0, 1$, then (β_1, θ_1) and (β_0, θ_0) are identified by the population regressions (7), and $\kappa(x) = W'(\theta_1 - \theta_0)$ is identified for every x in the support of X .*

Proof. Write $R_1 = (X', W'\lambda_1(Z\gamma))'$. Under A1–A2, $\mathbb{E}[Y \mid X, Z, D = 1] = R_1'(\beta'_1, \theta'_1)'$ exactly, so the coefficient of the population regression of Y on R_1 in the treated subpopulation is $(\beta'_1, \theta'_1)'$ whenever $\mathbb{E}[R_1 R_1' \mid D = 1]$ is nonsingular (γ is identified by the probit of D on Z under A1, and enters R_1 as a known function). Suppose $a'X + c'W\lambda_1(Z\gamma) = 0$ almost surely on $D = 1$ for some $(a, c) \neq 0$. Conditional on $X = x$, the left side is $a'x + c'W(x)\lambda_1(Z\gamma)$; by E and V, $Z\gamma$ takes at least two values with positive probability in the treated subpopulation at almost every x , and λ_1 is strictly decreasing, so $\lambda_1(Z\gamma)$ takes two distinct values there; hence $c'W(x) = 0$ and then $a'x = 0$ for almost every x . The first equality with $\mathbb{E}[WW' \mid D = 1]$ nonsingular gives $c = 0$; the second with $\mathbb{E}[XX' \mid D = 1]$ nonsingular gives $a = 0$, a contradiction. The untreated regime is identical with λ_0 , which is strictly increasing. The difference $\theta_1 - \theta_0$ is then identified, and $\kappa(x) = W(x)'(\theta_1 - \theta_0)$ with it. $\square$

Two remarks. First, the proof uses the exclusion restriction only through V: what identifies θ_j is that λ_j moves while X stays fixed. Without an excluded variable, $Z = X$, the same argument goes through with $\lambda_j(X\gamma)$ in place of $\lambda_j(Z\gamma)$ as long as $\lambda_j(X\gamma)W(X)$ is not a linear function of X ; this is identification by the curvature of λ_j alone, the case Section 3 treats as degenerate, and it is more fragile for $\kappa(x)$ than for κ because the interactions $W\lambda_j$ are closer to collinear with X than λ_j is. Second, the rank condition on W is the only addition: W must not contain redundant functions of X within each regime. In the leading case $W = (1, X_{-1})$ it

is the usual full-rank condition on the regressors. When the index varies with X held fixed only on a subset $\mathcal{A}$ of the support of X , the same argument identifies θ_j as long as $\mathbb{E}[WW'\mathbf{1}\{X \in \mathcal{A}\} \mid D = j]$ is nonsingular.

Reading $\kappa(x)$. The slopes in (8) are the paper’s answer to the question “who selects on gains?”. Under the targeting story of Araar (2026a), a program that reaches households whose return rises with endowments should show $\partial\kappa(x)/\partial w_k > 0$ for the endowment variables; a program taken up on the basis of private information that does not vary with endowments should show slopes near zero; a negative slope means that the households with more of w_k participate less on the basis of their gains, which is what one expects when w_k measures a constraint that the program relaxes for everyone. Section 4.3 gives the test of the constant model, Section 5 the two routes, and Section 6 the size and power of the test.

2.3 Assumptions, ordered by what they cost

The ESR is usually presented as one assumption, trivariate normality, and accepted or rejected as a block. The paper takes it apart. Three assumptions bear on the unobservables and three properties on the selection equation; they are listed here in the order in which the estimation routes of Section 5 need them, which is also the order in which they are worth testing.

Assumption 1 (A1, selection). u is standard normal and independent of Z , and participation depends on Z through the single index $Z\gamma$.

Assumption 2 (A2, linear conditional means). $\mathbb{E}[\omega_j \mid X, Z, u] = \rho_j \sigma_j(X) u$ for $j = 0, 1$.

Assumption 3 (A3, trivariate normality). (ω_1, ω_0, u) is jointly normal given X .

A1 is what makes λ_1 and λ_0 the right functions and $P(Z)$ a probit; it is the assumption on the selection equation’s *form*. A2 is what makes them the right *controls*: with A1 it gives the regime regressions their shape and identifies $\rho_j \sigma_j$ and hence κ , with no further assumption on the distribution of (ω_1, ω_0) ; A2 is written with $\rho_j \sigma_j(X)$ to accommodate Section 2.2. A3 implies A1 and A2, adds the normality of the regime errors around their conditional means, and buys the efficiency of the likelihood; it costs the consistency of the likelihood when it fails, while A2 without A3 costs nothing but efficiency. Three properties of the selection equation sit beside these:

- (E) *argument*: Z contains at least one variable excluded from X , with a non-zero coefficient in γ ;
- (S) *sufficiency*: $\mathbb{E}[\omega_j \mid X, Z, u] = \mathbb{E}[\omega_j \mid X, u]$, so that two values of Z with the same index carry the same distribution of the regime errors (Section 4.2);
- (V) *variation*: $\text{Var}(Z\gamma \mid X) > 0$ with positive probability in each regime, and the ranges of $P(Z)$ among the treated and the untreated overlap.

E is the exclusion restriction; S is what is implicitly assumed whenever a single index is written, and is stated here because it can fail on its own (a direct effect of an instrument, a second index in participation) and because it is testable; V is the strength and support of the instrument, a magnitude rather than a hypothesis. Table 1 lists what each assumption buys, which route needs it, and where it is checked.

The reading of the table is the argument of the paper in one line: the only assumption that can be dropped at no cost to identification is A3, by the two-step route; A2 can be weakened to smoothness by the semiparametric control in P , at the cost of the parametric line; A1, E, S and V cannot be weakened without losing the object, and are therefore the ones to check.

Table 1: Assumptions, what they buy, which route needs them, and where they are checked

	What it buys	Needed by	Checked by
A1	the probit for $P(Z)$; the Mills ratios as functions	all routes (form of λ_j , of the strata, of the Mills factors)	specification of the probit (§4.1); semiparametric curve vs line (§5.4)
A2	λ_j as sufficient control; identification of $\rho_j\sigma_j$, κ , $\kappa(x)$	two-step, likelihood, pseudo-DiD	Hermite controls, esrtest , normal (§4.5)
A3	efficiency of the likelihood; the line extrapolated in the tails	likelihood only	Hausman FIML vs two-step, esrtest , normal (§4.5)
E	identification by variation rather than by form	all routes (without it, identification by form)	esrdiag : LR of the excluded instruments (§4.4)
S	a single index carries all the selection information	all routes	esrtest , suff (§4.2)
V	precision of $\rho_j\sigma_j$; effects estimated rather than extrapolated	all routes	esrdiag : $\text{Var}(P X) / \text{Var}(P)$, VIF of λ_j , support and extrapolated shares (§4.4)

3 What each route identifies

Three routes estimate the model of Section 2, and they use the assumptions of Table 1 in a nested way.

Proposition 2 (Identification by route). *Let E , S and V hold.*

- (i) *Under A1–A3 the parameters $(\beta_1, \beta_0, \gamma, \sigma_1, \sigma_0, \rho_1, \rho_0)$ are identified by the likelihood of (Y, D) given Z , and with them κ , the effects and the MTE line (5) on the whole of $(0, 1)$.*
- (ii) *Under A1–A2 the parameters $(\gamma, \beta_1, \rho_1\sigma_1, \beta_0, \rho_0\sigma_0)$ are identified by the probit and the two regime regressions on X and $\lambda_j(Z\gamma)$ (Proposition 1 with $W = 1$); κ , the effects and the line (5) follow, but σ_j and ρ_j separately do not, and the line is trusted on the support of $P(Z)$ only.*
- (iii) *Under A1 alone, with $\mathbb{E}[Y | X, Z, D = j]$ a smooth function of $(X, Z\gamma)$, the functions $m_j(x, p) = \mathbb{E}[Y_j | X = x, P = p, D = j]$ are identified on the support of $P(Z)$, and with them the marginal treatment effect $\text{MTE}(x, p)$ on that support, by the derivative of $\mathbb{E}[Y | X, P]$ in P (Heckman and Vytlacil, 2005); κ is identified as the average slope of the MTE only if the slope is constant, that is under A2.*

Part (i) is the classical result and part (ii) is Proposition 1; part (iii) is the local instrumental variable identification of Heckman and Vytlacil (2005) restricted to the index model. The proposition orders the routes by what they return: the likelihood returns the most, including the shape of the line outside the support, and needs the most; the two-step returns the same objects on the support with A2 only; the semiparametric control returns a curve rather than a line and gives up κ as a single number unless the curve is straight. The practical consequence is that the three routes should agree when the model is right and disagree in a readable way when it is not, and Section 4 turns each disagreement into a test.

Identification by form. When E fails, $Z = X$, and V fails with it in the sense that $Z\gamma$ is a function of X . Nothing in Proposition 2 requires $\lambda_j(X\gamma)$ to be linearly independent of X as a matter of logic: λ_j is nonlinear, so the regime regressions remain of full rank and both parametric

routes return numbers. What is lost is the source of the identification. The coefficient of λ_j is then identified by the curvature of λ_j over the range of $X\gamma$, that is by the choice of the normal law in A1 and the linearity of X in the outcome equations; any misspecification of either loads directly on $\rho_j\sigma_j$, and the variance inflation of Section 4.4 says how much. The paper treats this case as degenerate: it is reported, with a warning, and read as a functional-form result rather than a causal one. The semiparametric route of part (iii) does not exist without E, since the derivative in P at fixed X is not defined when P is a function of X .

When S fails. The routes are not robust to a failure of index sufficiency, and the bias has a simple form on the two-step route. Suppose an excluded instrument z_2 enters the regime errors directly, $\omega_j = \zeta_j z_2 + \tilde{\omega}_j$ with $\tilde{\omega}_j$ satisfying A2 and S. The regression of Y on X and λ_j within regime j omits $\zeta_j z_2$, and the coefficient of λ_j converges to

$$\rho_j\sigma_j + \zeta_j \frac{\text{Cov}^*(z_2, \lambda_j \mid D = j)}{\text{Var}^*(\lambda_j \mid D = j)}, \quad (12)$$

where Cov^* and Var^* are taken after partialling X out. The second term does not vanish with n , it has the sign of $\zeta_j\gamma_{z_2}$ times the sign of the slope of λ_j (negative for λ_1 , positive for λ_0), and it enters κ as the difference of the two regime biases. The sufficiency test of Section 4.2 estimates ζ_j directly when a second instrument is available, and (12) says what a non-zero $\hat{\zeta}_j$ does to κ if the instrument is kept. Under a second index in participation the bias has no closed form, and the same test is the only guard.

4 Tests borne by the selection equation

The tests of this section follow the order of Table 1. Every one of them is carried by the selection equation: either it examines the probit itself (Sections 4.1 and 4.4), or it uses the index the probit produces to look at the outcomes (Sections 4.2, 4.3, 4.5 and 4.6). This is the reason for putting the selection equation first: what is wrong there is inherited by everything downstream.

4.1 Specification of the selection equation

Under Assumption A1 the participation index $Z\gamma$ is the only channel through which Z reaches the regimes, and everything the estimators use is a function of its estimate: the Mills ratios $\lambda_j(Z\hat{\gamma})$ that both routes take as control functions, the score $P(Z) = \Phi(Z\hat{\gamma})$ on which the MTE is drawn, the strata of the pseudo-DiD. A single index may well exist and still be misspecified in the working probit—a missing polynomial or interaction in Z , a link that is not the normal cdf—and then λ_j is the wrong function in every unit and both routes fit the wrong curve. This is a different question from the one of Section 4.2, which asks whether Z matters beyond the index; here the question is whether the index is the right function of Z . The two are complementary and the answer to the second is needed before the first is asked, since the sufficiency test adds instruments to regressions that already contain $\lambda_j(Z\hat{\gamma})$.

The link test. Let $\hat{v} = Z\hat{\gamma}$ be the fitted index of the probit. The link test of Pregibon (1980) refits the probit of D on $\hat{v}$ and $\hat{v}^2$ (and $\hat{v}^3$ when a third order is asked for): if the index is correctly specified, $\hat{v}$ carries all the information and the powers have zero coefficients, while the coefficient of $\hat{v}$ is one. The statistic is the Wald test of the powers, χ^2 with one degree of freedom per power. It is cheap, it needs the probit alone, and it rejects both a wrong functional form of the index and a wrong link; it does not say which, and it does not say which term is missing. The second form of the test does. With named candidates—squares, interactions, a spline of an instrument—the probit is refitted with the terms added to Z and their coefficients are tested, by Wald and by the likelihood ratio of the two probits. This is how the index is chosen in practice,

and the ESR puts no premium on a parsimonious selection equation: a term that belongs to Z and to X costs nothing, and a term added to Z but not to X is an exclusion restriction like any other, to be examined with the others in Section 4.2.

What trivariate normality imposes on the selection equation. The probit estimates γ from the choices alone and is consistent under A1. The likelihood estimates γ jointly with the outcome equations, and the outcomes carry information about γ through the correlations: the density of Y given D reads the index through λ_j , and the score of the likelihood with respect to γ has a component from each regime that vanishes only when $\rho_1 = \rho_0 = 0$. Under A1–A3 both estimators are consistent and the likelihood is efficient. When A3 fails, the likelihood moves γ to fit the regime distributions it is told are normal, and the probit does not move. When the index itself is misspecified, both are inconsistent but not in the same way: the probit fits the wrong index by its own criterion, the likelihood trades the fit of D against the fit of Y . Either way the two estimates part company, and the contrast

$$\mathcal{H}_\gamma = (\hat{\gamma}_P - \hat{\gamma}_F)' \hat{V}_{\Delta, \gamma}^+ (\hat{\gamma}_P - \hat{\gamma}_F) \quad (13)$$

is the γ block of the Hausman statistic of Section 4.5, with the same influence-function covariance (18) restricted to the selection coefficients. It is contained in the full contrast, and it is reported on its own for what it localizes: this is exactly what trivariate normality imposes on the selection equation, that the choices and the outcomes, read through the normal likelihood, agree about γ . A rejection of (13) together with a rejection of the link test points to the index; a rejection with a clean link test points to A3, and Section 4.5 takes over.

A single sample. The baseline design of Section 6 with a quadratic in the index, $D = \mathbf{1}\{0.2 + 0.3x + 0.9z + 0.5(z^2 - 1) + u > 0\}$, joint normal errors and $\kappa = 1.08$, $n = 5,000$ (file `esr.link`). The working probit on (x, z) gives a link statistic of 307.9 on one degree of freedom, and the contrast (13) rejects at 31.2 on three; adding z^2 to the probit, the coefficient of z^2 is 0.494 (0.023) and the two statistics fall to 0.95 and 2.64 (on four degrees of freedom, $p = 0.62$). The cost of the wrong index on the parameter of interest is moderate here, $\hat{\kappa} = 0.93$ (two-step) and 0.98 (FIML) against 1.13 and 1.16 with the quadratic, about one and a half standard errors; it grows with the curvature the index omits and with the correlations, and nothing in the output of the misspecified model signals it. On the baseline sample `esr.sim`, where the index is right, the link statistic is 0.05 and the contrast 6.7 on three degrees of freedom ($p = 0.08$). The Monte Carlo of Section 6 gives the size and power of both. In the command, `esrttest, spec` runs the link test (option `order(#)`) or the test of added terms (option `add(varlist)`) and the contrast (13).

4.2 Index sufficiency

The single-index selection equation says more than that D depends on Z through $Z\gamma$: it says that, given X , the regime errors depend on Z through $Z\gamma$ only, through the participation error u . Two vectors z and z' with the same index have the same probability of participation, and Assumption S requires that they also have the same conditional distribution of (ω_1, ω_0) among participants and among non-participants:

$$(S) \quad \mathbb{E}[\omega_j \mid X, Z, u] = \mathbb{E}[\omega_j \mid X, u], \quad j = 0, 1. \quad (14)$$

This is what makes the control function $\lambda_j(Z\gamma)$ sufficient. It is distinct from the exclusion restriction, which is about the outcome equations, and it is usually left implicit; it fails in two ways that matter in practice. The first is a direct effect of an instrument on the outcome, $\omega_j = \zeta_j z_2 + \tilde{\omega}_j$: the exclusion of z_2 is wrong, and the regime errors depend on z_2 beyond the index. The second is a second index in participation: when the decision has two hurdles, or two

unobservables that different instruments move, the probability of participation is still a function of Z , but the composition of the participants at a given probability depends on which instrument brought them in, and $\mathbb{E}[\omega_j \mid X, Z, D]$ is no longer a function of $Z\gamma$ alone. In both cases the ESR estimates, κ first among them, are biased, and the bias does not shrink with the sample size.

The test. Under A1–A2 and S, adding any function of Z to the regime regressions (7) beside X and $\lambda_j(Z\gamma)$ leaves its coefficient at zero. The test adds the excluded instruments themselves, linearly, with one exception: one excluded instrument, z_{keep} , is held out so that λ_j keeps a source of variation that is not collinear with the added terms. Let Z_2 be the excluded instruments, Z_2^- the vector without z_{keep} , and consider the two regressions

$$Y = X\beta_1 + Z_2^- \pi_1 + \theta_1 \lambda_1(Z\hat{\gamma}) + e_1 \quad (D = 1), \quad Y = X\beta_0 + Z_2^- \pi_0 - \theta_0 \lambda_0(Z\hat{\gamma}) + e_0 \quad (D = 0). \quad (15)$$

The null is $\pi_1 = \pi_0 = 0$, and the statistic is the Wald form $\mathcal{W}_S = \hat{\pi}'(\hat{V}_\pi)^{-1}\hat{\pi} \rightarrow \chi^2(2 \dim Z_2^-)$ with $\hat{V}_\pi$ from the stacked moment conditions of the whole procedure, which account for the estimation of γ in the first step. The test is also reported regime by regime. With a single excluded instrument Z_2^- is empty; the command then adds the instrument itself, and the test rests on the curvature of λ_j alone, that is on identification by form (Section 3); it is reported with a warning, and it is the case where the practitioner has no way to separate a failure of S from a failure of the functional form. The choice of z_{keep} is the strongest instrument in the probit by default, and can be imposed. What the test cannot see is a direct effect of z_{keep} itself: held out so that λ_j stays identified, it is never added, and a direct effect of it on the outcome goes into θ_j and κ unseen. With two or more excluded instruments, rotating z_{keep} (the option `keep()`) and reporting each version covers every instrument in turn; with one, its exclusion is an assumption that no test of the family can check.

Proposition 3 (The sufficiency test). *Under A1, A2, E, V and S, $\pi_1 = \pi_0 = 0$ in the population regressions (15), and $\mathcal{W}_S \rightarrow \chi^2(2 \dim Z_2^-)$. If S fails through a direct effect $\omega_j = \zeta_j z_2 + \tilde{\omega}_j$ with $\tilde{\omega}_j$ satisfying S, then $\pi_j = \zeta_j$: the regressions (15) estimate the direct effect, and the test is consistent against it.*

Proof. Under A1–A2 and S, $\mathbb{E}[Y \mid X, Z, D = 1] = X\beta_1 + \rho_1 \sigma_1 \lambda_1(Z\gamma)$ exactly, so the population coefficient of Z_2^- in the regression of Y on $(X, Z_2^-, \lambda_1(Z\gamma))$ is zero provided the design has full rank, which holds when $\lambda_1(Z\gamma)$ varies at fixed (X, Z_2^-) , that is when z_{keep} has a non-zero coefficient in γ and varies conditionally on (X, Z_2^-) (E and V). The first-step estimation of γ is a smooth transformation and the stacked moment-condition variance is the delta-method variance of the two-step estimator, so the Wald statistic has its χ^2 limit under the null. Under the direct-effect alternative, $\mathbb{E}[Y \mid X, Z, D = 1] = X\beta_1 + \zeta_1 z_2 + \rho_1 \sigma_1 \lambda_1(Z\gamma)$ by the same argument applied to $\tilde{\omega}_1$, so $\pi_1 = \zeta_1$; the untreated regime is identical. $\square$

Against a second index the test has no closed-form alternative, but the mechanism is clear: with two hurdles the treated are those who passed both, $\mathbb{E}[\omega_j \mid X, Z, D = 1]$ depends on the two indices separately, and the single-index $\lambda_1(Z\gamma)$ leaves a residual dependence on each instrument that the linear term $Z_2^- \pi_1$ picks up. The Monte Carlo of Section 6 makes this concrete on three designs with two instruments. When S holds the rejection rate at the 5 percent level is 0.045 ($n = 2,000$) and 0.035 ($n = 5,000$). When z_2 enters the outcomes with coefficient -0.3 the test rejects always, and $\hat{\pi}_1$ and $\hat{\pi}_0$ average -0.30 in both regimes, the direct effect. Under a double hurdle in which the regime errors are tied to the first hurdle only, the joint test rejects in 93 and 100 percent of the samples, almost entirely through the treated regime, where the added instrument takes a coefficient of 0.24: the untreated pool mixes those who failed either hurdle and its composition is less sensitive to the second index. Finally, on a design where S holds but $\mathbb{E}[\omega_1 \mid u]$ is cubic in u , so that A2 fails, the rejection rate stays at 0.035 and 0.07: a linear term in the instrument does not absorb a nonlinearity in u , and the sufficiency test is not confounded

with the normality test of Section 4.5. The command `esrtest, suff [keep()]` computes (15) after any `esreg` estimation and reports $\hat{\pi}_j$, the joint and the by-regime statistics.

The test is the parametric, control-function version of the test of linearity in P of Heckman and Vytlačil (2005): under the index model $\mathbb{E}[Y \mid X, Z]$ depends on Z through $P(Z)$ only, and adding Z beside functions of P tests the same restriction without A2. The version here uses λ_j as the control and tests within regimes, so it needs A2 and gains power from it; the semiparametric version of Section 5.4 is the natural check when A2 is in doubt.

4.3 Constant κ in X

The constant model of Araar (2026a) is the restriction $\theta_{1,-1} = \theta_{0,-1}$ in the two-step form (8), a set of $p_w - 1$ linear restrictions on the coefficients of the interactions $W\lambda_j$ across the two regimes. The test statistic is the Wald form

$$\mathcal{W}_\kappa = (R\hat{\vartheta})'(R\hat{V}R')^{-1}(R\hat{\vartheta}) \xrightarrow{d} \chi^2(p_w - 1) \quad \text{under } H_0 : \theta_{1,-1} = \theta_{0,-1}, \quad (16)$$

where $\hat{\vartheta} = (\hat{\gamma}, \hat{\beta}_1, \hat{\theta}_1, \hat{\beta}_0, \hat{\theta}_0)$ is the two-step estimator, R selects $\theta_{1,-1} - \theta_{0,-1}$, and $\hat{V}$ is the covariance of the whole two-step procedure from the stacked moment conditions of Section 5.2. Using $\hat{V}$ rather than the regression-by-regression covariances matters here: the two restrictions involve coefficients estimated on disjoint subsamples but with a common first step, and the estimation of γ moves λ_1 and λ_0 in opposite directions. The null is exact, in the sense that H_0 is the constant model and nothing else, and the test needs A1–A2 only.

In the likelihood form the constant model is $\alpha_{j,-1} = \delta_{j,-1} = 0$, $j = 0, 1$, and the Wald test of these $2(p_s - 1) + 2(p_r - 1)$ restrictions on the FIML estimates is a test of homogeneous regime laws, a sufficient condition for $\kappa(x)$ constant but not a necessary one, since $\kappa(x)$ can be constant with $\sigma_1(x)$ and $\rho_1(x)$ both varying. It is the natural test when the FIML route is retained, and it has more power than (16) against alternatives where the whole law moves with x ; it also inherits the fragility of the likelihood under departures from A3, so that a rejection after the FIML fit should be read together with the normality test of Section 4.5.

A stratified version of either test replaces W by indicators of G strata of X and tests the equality of the G within-stratum values of κ ; it is the version to use when the heterogeneity is expected in groups rather than along a continuous variable, and when the strata are large enough for the selection equation to be estimated within each. The Monte Carlo of Section 6 reports the size and power of (16) on the design of Appendix B with $\kappa(x) = \kappa + sx$: with $n = 5,000$ the rejection rate at the 5 percent level is 0.045 under $s = 0$ and 0.29, 0.72 and 0.995 under $s = 0.1, 0.2$ and 0.4 , and the slope estimate is unbiased with a standard deviation of 0.08; with $n = 2,000$ the size is 0.03 and the power 0.35 and 0.89 at $s = 0.2$ and 0.4 . The command `esrtest, kappa` computes (16) after `esreg, method(twostep) kappa()` and its likelihood counterpart after `esreg, hetsigma() hetro()`, and reports each slope $\hat{\theta}_{1k} - \hat{\theta}_{0k}$ with its standard error.

A caveat that the next subsections make precise: the interactions $W\lambda_j$ also absorb two other departures from the model, a second index in the selection equation (Section 4.2) and a nonlinear conditional mean of ω_j in u (Section 4.5). A rejection of the constant model is evidence against the constant model, not by itself evidence for heterogeneous selection on gains; the reading is secured when the sufficiency and normality tests do not reject.

4.4 Strength and support

The third property of the selection equation is its variation: how much the score $P(Z)$ moves once X is held fixed, and over which range. The first governs the precision of everything that depends on λ_j , and the second the range over which the effects are estimated rather than extrapolated. Neither is a hypothesis to test; both are magnitudes to report, and `esrdiag` reports them after every `esreg` estimation.

Strength. In the two-step regression of regime j , the coefficient $\theta_j = \rho_j \sigma_j$ is the coefficient of $\lambda_j(Z\hat{\gamma})$ in a regression that also contains X . By the Frisch–Waugh theorem its conditional variance, given the first step, is

$$\text{Var}(\hat{\theta}_j \mid \hat{\gamma}) = \frac{\sigma_{e_j}^2}{n_j \widehat{\text{Var}}(\lambda_j) (1 - R_{\lambda_j \cdot X}^2)}, \quad (17)$$

where $R_{\lambda_j \cdot X}^2$ is the R^2 of the regression of $\lambda_j(Z\hat{\gamma})$ on X in regime j . The factor $\text{VIF}_j = (1 - R_{\lambda_j \cdot X}^2)^{-1}$ is the variance inflation of the Mills ratio, and it is where the strength of the instrument acts. When the excluded instruments are strong, $Z\hat{\gamma}$ varies widely at fixed X and λ_j is far from any linear function of X : VIF_j is close to one. When they are weak, $Z\hat{\gamma}$ is nearly $X\hat{\gamma}_x$ and λ_j is nearly a smooth function of the index $X\hat{\gamma}_x$ alone; its only variation independent of X is then the curvature of λ_j , which a linear X cannot absorb, and VIF_j grows without bound as the instrument's coefficient goes to zero. This is identification by form, and (17) says what it costs: the standard error of κ grows like $\sqrt{\text{VIF}_j}$. The likelihood route is not immune; it uses the same information, with the normality of the regime errors as an extra source, and its estimates become unstable rather than merely imprecise when the instrument disappears.

Three numbers summarize the strength and are reported together. The likelihood-ratio statistic of the excluded instruments in the probit, $\chi^2(\dim Z_2)$, with the increment in pseudo- R^2 they bring and the $\chi^2(1)$ of each instrument, is the analogue of the first-stage F . The share of the variance of the score not explained by X , $1 - R_{P \cdot X}^2$, is the analogue of the partial R^2 of the excluded instruments, on the scale of P . And VIF_1 , VIF_0 measure directly the collinearity that (17) penalizes. The Monte Carlo of Section 6 runs the baseline design with the instrument's coefficient γ_z at 0.9, 0.45, 0.2 and 0, $n = 5,000$: the LR statistic falls from 1,626 to 542, 118 and 1; $1 - R_{P \cdot X}^2$ from 0.90 to 0.69, 0.31 and 0.007; the VIF of λ_1 rises from 1.1 to 1.4, 3.1 and 96; and the standard deviation of $\hat{\kappa}$ over replications goes from 0.088 to 0.14, 0.30 and 1.9 on the two-step route, from 0.075 to 0.12, 0.21 and 0.63 on the likelihood route, where the root mean squared error at $\gamma_z = 0$ is 0.89: the likelihood still returns a number, and it is wrong by almost the size of the parameter. The rule of thumb the command applies is the one of the linear IV literature transposed: an LR statistic below 10 or a VIF above 10 is flagged, and the second flag is worded as what it is, identification by the curvature of λ alone.

Support. The ESR reports the ATU as the average of $X(\beta_1 - \beta_0) - \kappa \lambda_0(Z\gamma)$ over the untreated, and the ATT as the average of $X(\beta_1 - \beta_0) + \kappa \lambda_1(Z\gamma)$ over the treated. Both averages run over the whole group, including the untreated whose score lies below the smallest score observed among the treated, and the treated whose score lies above the largest score among the untreated. For those individuals the counterfactual is the parametric line evaluated where no observation of the other regime exists; it is an extrapolation, correct if the model is, and unchecked by the data. The common support $[p, \bar{p}]$ is the intersection of the two ranges of $P(Z)$, and `esrdiag` reports the weighted share of each group outside the other's range, together with the ATT and the ATU recomputed on the common support alone. The difference between the full and the restricted effect is the contribution of the extrapolated individuals; when it is small the question is moot, and when it is not the restricted effect is the one that the semiparametric route of Section 5.4 can also deliver, and the two should agree. On the baseline design the shares are 2.4 percent on either side and the restricted effects differ from the full ones by less than 0.005; on a design with a weak instrument the shares shrink, since a weak instrument keeps both groups on the same narrow range of scores, which is exactly why the support is not a substitute for strength.

The three ingredients, strength, variation and support, are the ones the practitioner should read before the effects, and they are the ones the road map of [Araar \(2026a\)](#) put first. The command prints them in that order with the two warnings, and returns them for the tables.

4.5 Normality by regime

Two assumptions remain once the selection equation has been checked, and they are the two that the likelihood route adds to the two-step route. Assumption A2 says that the conditional mean of each regime error given the participation error is linear, $\mathbb{E}[\omega_j | u] = \rho_j \sigma_j u$; it is what makes the Mills ratio the right control function, and both routes need it. Assumption A3 says that (ω_1, ω_0, u) is trivariate normal; it is what the likelihood uses beyond A2, and what the two-step does not need. [Araar \(2026b\)](#) showed on single samples that the two routes agree when A3 holds and part company when the regime errors are skewed, the likelihood being the one that moves. This section turns that observation into two tests, one for each assumption, and shows that they do not confound each other.

A3: the FIML against two-step contrast. Let $q = (\gamma, \beta_1, \rho_1 \sigma_1, \beta_0, \rho_0 \sigma_0)$ be the parameters that both routes estimate. Under A1–A3 both estimators are consistent and the likelihood is efficient; under A1–A2 with A3 false the two-step remains consistent and the likelihood does not. The Hausman principle applies, with one adjustment. The usual covariance of the difference, $V_{2S} - V_F$, rests on the efficiency of the likelihood under the null and is often not positive definite in finite samples; the paper uses instead the covariance built from the influence functions of the two estimators,

$$\hat{q}_{2S} - \hat{q}_F \approx \sum_i (\psi_{2S,i} - \psi_{F,i}), \quad \hat{V}_\Delta = \sum_i (\psi_{2S,i} - \psi_{F,i})(\psi_{2S,i} - \psi_{F,i})', \quad (18)$$

with $\psi_{F,i} = \hat{I}^{-1} s_i(\hat{\theta}_F)$ the score of observation i premultiplied by the inverse observed information of the sample (whatever variance the estimation reported), mapped to q by the delta method for $\rho_j \sigma_j = \tanh(c_j) \exp(a_j)$, and $\psi_{2S,i} = -\hat{G}^{-1} g_i(\hat{\vartheta})$ the contribution of observation i to the stacked moment conditions of the two-step, premultiplied by the inverse Jacobian. $\hat{V}_\Delta$ is positive semi-definite by construction, equals $V_{2S} - V_F$ in large samples under the null, and remains a valid covariance of the difference under the alternative, so that the contrast can be reported with a standard error whether or not it rejects. The statistic is $\mathcal{H} = (\hat{q}_{2S} - \hat{q}_F)' \hat{V}_\Delta^+ (\hat{q}_{2S} - \hat{q}_F)$, with χ^2 degrees of freedom equal to the rank of $\hat{V}_\Delta$. The one-dimensional contrast on κ , $(\hat{\kappa}_{2S} - \hat{\kappa}_F)/\widehat{\text{se}}$, is reported beside it: it is the number a practitioner reads, and it is the difference that [Araar \(2026b\)](#) displayed without a standard error.

A2: Hermite controls. If $\mathbb{E}[\omega_j | u]$ is not linear but a polynomial in u , the control function is not λ_j alone. Writing $\mathbb{E}[\omega_j | u] = a_{j1}u + a_{j2}(u^2 - 1) + a_{j3}(u^3 - 3u)$ in the Hermite basis, the conditional means given participation are, for $u \sim N(0, 1)$,

$$\begin{aligned} \mathbb{E}[Y | X, Z, D = 1] &= X\beta_1 + a_{11}\lambda_1 - a_{12}Z\gamma\lambda_1 + a_{13}((Z\gamma)^2 - 1)\lambda_1, \\ \mathbb{E}[Y | X, Z, D = 0] &= X\beta_0 - a_{01}\lambda_0 + a_{02}Z\gamma\lambda_0 - a_{03}((Z\gamma)^2 - 1)\lambda_0, \end{aligned} \quad (19)$$

since $\mathbb{E}[u^2 - 1 | u > -v] = -v\lambda_1(v)$ and $\mathbb{E}[u^3 - 3u | u > -v] = (v^2 - 1)\lambda_1(v)$, with the mirror expressions below the threshold. The two extra terms are known functions of the index and enter each regime regression beside λ_j ; under A2 their coefficients a_{j2}, a_{j3} are zero, and the Wald statistic on the four of them, with the stacked variance, is $\chi^2(4)$ under the null. The test needs A1 only, and it targets the shape of the dependence on u , not the law of the regime errors around it: a skewed ω_j with a linear conditional mean leaves $a_{j2} = a_{j3} = 0$.

Proposition 4 (Separation of the two tests). *Under A1–A2 and E, V, the Hermite coefficients in (19) are zero whether or not A3 holds, so that the Hermite test has its nominal size under skewed regime errors; under A1–A3 the Hausman contrast (18) has its χ^2 limit. Under a failure of A3 that keeps A2 (skewed regime errors with a linear conditional mean), the likelihood is inconsistent for q and the Hausman contrast rejects, while the Hermite test does not. Under*

a failure of A2 (a nonlinear conditional mean), the Hermite test rejects, and both routes are inconsistent for $\rho_j\sigma_j$ and κ .

The proof is the computation above for the Hermite terms, the consistency of the two-step under A1–A2 for the Hausman part (Proposition 1 with $W = 1$), and the inconsistency of the likelihood under a wrong joint law, which is the content of the proposition of Araar (2026b). The last statement of the proposition is the important one for practice: when A2 fails, no parametric line is right. The augmented two-step of Section 5.3 then keeps the Hermite terms in the estimation, when the instrument is strong enough to separate them from the Mills ratio, and the semiparametric curve of Section 5.4 is its check, or the estimate of the marginal treatment effect to report when it is not.

The Monte Carlo of Section 6 runs both tests on the three designs of Araar (2026a) with $n = 5,000$: N (joint normal), C2 (skewed regime errors, A2 holds, A3 fails) and C1 (gain nonlinear in u , A2 fails). Under N the Hausman contrast rejects in 2 percent of the samples and the κ contrast in 7; the Hermite test in 7 percent, and in 4.5 to 7.5 percent over $n = 2,000$ and 5,000 with 200 replications. Under C2 the Hausman contrast rejects always, with $\hat{\kappa}_{2S} - \hat{\kappa}_F$ averaging -1.8 , and the Hermite test in 7 percent: the skewness moves the likelihood and leaves the shape alone. Under C1 the Hermite test rejects in 94 percent of the samples (through the treated regime, where the gain bends), the Hausman contrast in 52: both routes are wrong in the same direction and the contrast is weak evidence, as the proposition says. The command `esrtest`, `normal` re-estimates both routes on the sample of the current `esreg` estimation and reports the two blocks, the contrast on each element of q and on κ , and the Hermite coefficients by regime, with the Wald statistics jointly, by regime and for the two cubic terms; the last two guide the augmented two-step (Section 5.3).

4.6 A model-light test of κ : the pseudo-DiD on the index

The tests above check the assumptions under which the ESR identifies κ . This last one turns the question around and asks what the data say about selection on gains with as little of the model as possible. It is the within-group comparison of Araar (2026a, Proposition 4) written as a difference-in-differences table, and it is the device a practitioner can put on one page next to the descriptive statistics.

The table. Estimate the selection equation by probit and let $v = Z\hat{\gamma}$ be the index. Cut the sample into G strata of v (two halves by default), and within each stratum s regress Y on X and D : the coefficient of D is the X -adjusted gap Δ_s between treated and untreated at that level of the propensity. Under A1–A2,

$$\Delta_s = \bar{x}_s(\beta_1 - \beta_0) + \rho_1\sigma_1\bar{\lambda}_{1,s} + \rho_0\sigma_0\bar{\lambda}_{0,s}, \quad (20)$$

where $\bar{\lambda}_{1,s}$ is the mean of $\lambda_1(v)$ among the treated of stratum s and $\bar{\lambda}_{0,s}$ the mean of $\lambda_0(v)$ among its untreated. Since λ_1 decreases and λ_0 increases in v , the gap moves across the strata whenever $\rho_1\sigma_1 \neq 0$ or $\rho_0\sigma_0 \neq 0$: with $\kappa > 0$ in the usual configuration ($\rho_1\sigma_1 > 0$, $\rho_0\sigma_0 < 0$) the gap is largest in the bottom stratum, where the treated are the eager few whose unobservables pushed them in and the untreated are the many with ordinary unobservables, and smallest at the top, where the untreated are the reluctant few. The double difference of the gaps, top minus bottom, is $\rho_1\sigma_1\Delta\bar{\lambda}_1 + \rho_0\sigma_0\Delta\bar{\lambda}_0$ with $\Delta\bar{\lambda}_1 < 0 < \Delta\bar{\lambda}_0$, a negative number under selection on gains; it is not κ times a constant unless the two Mills contrasts happen to be equal in absolute value, which is why the table is read through its two within-group pieces rather than through the double difference alone.

The two contrasts. Within the treated, regress Y on X and the indicator of the top stratum over the units of the two extreme strata: the coefficient of the indicator is the X -adjusted

contrast C_1 ; within the untreated, the same regression gives C_0 . The same two regressions with the Mills ratio $\lambda_j(v)$ in place of the outcome give its X -adjusted contrasts, $\pi_1 < 0$ among the treated and $\pi_0 > 0$ among the untreated. Under A1–A2, $C_1 = \rho_1\sigma_1\pi_1$ and $C_0 = -\rho_0\sigma_0\pi_0$ in the population, exactly, since $X\beta_j$ is absorbed by the regressors. Two things follow. First, the signs of C_1 and C_0 identify the signs of $\rho_1\sigma_1$ and $\rho_0\sigma_0$ without A1 or A3: all that is needed is A2 and the monotonicity of $\mathbb{E}[u \mid u > -v]$ and $\mathbb{E}[u \mid u < -v]$ in v , which holds for any continuous law of u . A negative C_1 says that the treated who entered against a low propensity have higher outcomes than the treated who entered with a high one, selection on gains among participants; a positive C_0 says that the untreated who stayed out despite a high propensity have higher untreated outcomes, the mirror pattern. Second, dividing by the Mills contrasts, which A1 makes computable from the probit, gives

$$\hat{\kappa}_{DD} = \frac{C_1}{\pi_1} + \frac{C_0}{\pi_0}, \quad \widehat{\rho_1\sigma_1} = \frac{C_1}{\pi_1}, \quad \widehat{\rho_0\sigma_0} = -\frac{C_0}{\pi_0}, \quad (21)$$

an estimator of κ that uses A1–A2 and nothing about the joint law of the regime errors. Each ratio is an instrumental-variables estimate of the coefficient of λ_j in the regime regression, with the stratum indicator as the instrument. The Mills contrast is adjusted for X as the outcome contrast is: X and the index are correlated whenever X enters Z , and a raw contrast of mean Mills ratios would then be the wrong denominator. The estimator is less efficient than the two-step by construction, and its virtue is that every number in it can be read off four regressions within groups.

Standard errors. The estimating equations of the whole procedure are stacked, and they are just identified: the probit score, the weighted shares of the index below the cut points that define the strata, the regressions of the gap table, and the four contrast regressions of Y and λ_j on X and the top indicator. The influence function of each estimate is the corresponding row of $-\hat{J}^{-1}m_i$, with m_i the stacked equations of unit i and $\hat{J}$ their Jacobian, the derivatives in γ and in the cut points taken on kernel-smoothed moments since the indicators of the strata are not differentiable; it is aggregated as the estimation was, by observation, by cluster or through the survey design. Two cases are flagged: a boundary between strata on a block of tied values of the index, which a resample moves as a block, and Mills contrasts estimated with a coefficient of variation above ten percent, which make the ratios far from normal. In both the standard errors are approximate, the signs of C_1 and C_0 remain readable, and the option `reps(#)` gives the bootstrap of the whole procedure.

Proposition 5 (Pseudo-DiD). *Under A1, A2, E and V, $C_1 \rightarrow \rho_1\sigma_1\pi_1$ and $C_0 \rightarrow -\rho_0\sigma_0\pi_0$, so that $\hat{\kappa}_{DD} \rightarrow \kappa$ for any fixed number of strata. Under A2 alone with u continuous, C_1 and C_0 are $\rho_1\sigma_1$ and $\rho_0\sigma_0$ times the X -adjusted contrasts of $\mathbb{E}[u \mid u > -v]$ and $\mathbb{E}[u \mid u < -v]$, two decreasing functions of v ; when the adjustment for X preserves the ordering of the two strata, $\text{sign}(C_1) = -\text{sign}(\rho_1\sigma_1)$ and $\text{sign}(C_0) = -\text{sign}(\rho_0\sigma_0)$.*

The first part holds because the conditional mean of Y in each group is $X\beta_j$ plus $\pm\rho_j\sigma_j\lambda_j(v)$, and X is among the regressors; the second uses only that $\mathbb{E}[\omega_j \mid D, v]$ is $\rho_j\sigma_j$ times a monotone function of v under A2, whatever the law of u . The adjustment for X within strata matters: without it the strata differ in $\bar{x}_s$ and the contrasts pick up $\bar{x}_s(\beta_1 - \beta_0)$; with a rich X the regression adjustment can be replaced by matching on X within strata. One thing must never be done: the strata must be cut on the index, not on the outcome. Splitting on Y conditions on a collider, and the resulting contrast of dispersions belongs to quantile treatment effects, not to selection on gains.

Under the homogeneous design H of Araar (2026a), where $\kappa = 0$ but $\rho_1\sigma_1 = \rho_0\sigma_0 = 0.5$, the proposition gives $C_1 < 0$ and $C_0 < 0$: a common selection without selection on gains, a pattern the table shows and κ alone does not. The command `esrtest, pdid [nq()]` prints the table,

the double difference, the two contrasts and the two Mills contrasts with their standard errors, $\widehat{\rho_j\sigma_j}$, and $\hat{\kappa}_{DD}$ next to the **esreg** value. Its estimates and standard errors were checked against an independent implementation in the reference library, against the identities of the cluster and design aggregations, and against the bootstrap of the whole procedure (Section 8.4).

5 Estimation by tiers

The command **esreg** estimates the model by tiers, each tier dropping one assumption of Table 1. The two parametric tiers are implemented in the current version and validated against **movestay** and **msat** to every printed digit on the samples of [Araar \(2026b\)](#); the third tier is the semiparametric curve of **esrmte**; the fourth is described for completeness.

5.1 Full information

Under A1–A3 the contribution of observation i to the log-likelihood is, with $e_{ji} = (y_i - x_i\beta_j)/\sigma_j$ and $a_{ji} = (z_i\gamma + \rho_j e_{ji})/\sqrt{1 - \rho_j^2}$,

$$\ell_i = D_i [\ln \phi(e_{1i}) - \ln \sigma_1 + \ln \Phi(a_{1i})] + (1 - D_i) [\ln \phi(e_{0i}) - \ln \sigma_0 + \ln \Phi(-a_{0i})], \quad (22)$$

the likelihood of [Maddala \(1983, p. 224\)](#) and [Lokshin and Sajaia \(2004\)](#), the one **etregress**, **poutcomes** maximizes when the treatment is interacted with every outcome covariate, parameterized by $\ln \sigma_j = W_s \alpha_j$ and $\operatorname{atanh} \rho_j = W_r \delta_j$ so that the heterogeneous model of Section 2.2 is the same likelihood with more columns. The score is analytic in every parameter, including the heterogeneity coefficients, and is maximized by Stata’s **ml** with the observed information as the default variance; **vce(robust)** and **vce(cluster)** are available, the first being the sandwich that is also the right variance when A3 is in doubt but the estimate is kept. The command reports the likelihood-ratio statistic of $\rho_1 = \rho_0 = 0$, that is of no selection on unobservables at all, with the correct two degrees of freedom (or $2p_r$ under **hetrho()**); the independent-equations log-likelihood is computed in closed form from the probit and the two regime regressions, so that no second maximization is needed. Starting values come from the probit and the OLS regressions by regime, and the maximization converges in four to eight iterations on the validation samples, including the skewed one where the maximum is far from the truth.

5.2 Two steps with a stacked variance

Under A1–A2 the two-step estimator is the probit for γ followed by the two regressions

$$y_i = x_i\beta_1 + W_i\theta_1 \lambda_1(z_i\hat{\gamma}) + e_{1i} \quad (D_i = 1), \quad y_i = x_i\beta_0 - W_i\theta_0 \lambda_0(z_i\hat{\gamma}) + e_{0i} \quad (D_i = 0). \quad (23)$$

Its variance is usually obtained by bootstrap, as **msat** does, because the regression-by-regime standard errors ignore the first step and the heteroskedasticity that the truncation induces. The command computes it instead from the stacked moment conditions of the whole procedure. Let $\vartheta = (\gamma, \beta_1, \theta_1, \beta_0, \theta_0)$ and

$$g_i(\vartheta) = \left(s_i(\gamma)', D_i r'_{1i}(y_i - r_{1i}(\beta'_1, \theta'_1)'), (1 - D_i) r'_{0i}(y_i - r_{0i}(\beta'_0, \theta'_0)') \right)', \quad (24)$$

with $s_i(\gamma)$ the probit score and $r_{ji} = (x_i, W_i\lambda_j(z_i\gamma))$ the regime- j regressor vector, which depends on γ . The two-step estimator is the exactly identified GMM estimator solving $\sum_i g_i(\hat{\vartheta}) = 0$, and its asymptotic variance is the sandwich

$$\hat{V} = \hat{G}^{-1} \left(\sum_i g_i g_i' \right) \hat{G}^{-1'}, \quad \hat{G} = \frac{\partial \sum_i g_i(\vartheta)}{\partial \vartheta'} \Big|_{\hat{\vartheta}}, \quad (25)$$

where the off-diagonal blocks of $\hat{G}$ carry the derivative of $\lambda_j(z_i\gamma)$ with respect to γ , that is the first-step correction, and the outer-product term carries the heteroskedasticity.

Proposition 6 (Variance of the two-step estimator). *Under A1–A2, E , S , V and standard regularity conditions, $\sqrt{n}(\hat{\vartheta} - \vartheta) \rightarrow N(0, G^{-1}\Omega G^{-1'})$ with $G = \mathbb{E}[\partial g_i / \partial \vartheta']$ and $\Omega = \mathbb{E}[g_i g_i']$, and (25) is consistent for this variance. The delta method applied to $\hat{V}$ gives the variance of $\kappa = \theta_1 - \theta_0$ (constant model) and of $\kappa(x)$.*

The proposition is the M-estimation result for a two-step estimator written as one system (Newey, 1984; Wooldridge, 2010); the only point specific to the ESR is that the same γ enters both regime blocks through λ_1 and λ_0 , which the stacking handles without further work. The control-function estimator of `etregress`, `cfunction` uses the same device with a longer moment vector, since it solves for $\ln \sigma_j$ and $\text{atanh } \rho_j$ from variance moments as well and reports the sandwich as its only variance; the two-step of this paper keeps the regression-by-regime estimator, which is the one whose contrast with the likelihood (Section 4.5) is informative, and recovers σ_j from the residuals. On the skewed sample of Araar (2026b), where the note reports bootstrap standard errors of 0.054, 0.075, 0.045 and 0.074 for the ATT, ATU, ATE and κ from 100 replications, the influence functions below give 0.052, 0.073, 0.045 and 0.067, within bootstrap noise and in a fraction of a second. The stacked variance is also what makes the tests of Section 4 exact rather than approximate: the Wald statistics on $\theta_1 - \theta_0$, on the added instruments, on the Hermite controls, and the influence functions of the Hausman contrast all read it.

The effects. The ATT, ATU, ATE and the MTE line are averages over the units of functions of ϑ and of the data, so that their estimation error has two parts: that of $\hat{\vartheta}$, and that of the average itself. With $\psi_i = -\hat{G}^{-1}g_i(\hat{\vartheta})$ the influence of unit i on $\hat{\vartheta}$, an effect $\hat{\tau} = \sum_i w_i \tau_i(\hat{\vartheta}) / \sum_i w_i$ has the influence function

$$\phi_i = J\psi_i + \frac{w_i\{\tau_i(\hat{\vartheta}) - \hat{\tau}\}}{\sum_k w_k}, \quad J = \frac{\partial \hat{\tau}}{\partial \vartheta'}, \quad (26)$$

and the variance $\sum_i \phi_i^2$, which adds to the delta-method term the sampling variance of the average and twice the covariance of the two: the covariance is not zero, since both parts depend on (D_i, Z_i) . The same holds for the likelihood route with ψ_i the score of unit i premultiplied by the inverse observed information. The influence functions are aggregated as the estimation was: summed within clusters with the factor $G/(G - 1)$ for `vce(cluster)`, and through the strata, primary units and weights of the design for `vce(svy)` (and after the `svy` prefix), which also gives the two-step its cluster and design variances.

5.3 The augmented two-step

When the Hermite controls reject (Section 4.5), A2 fails and the line of both routes has the wrong shape. The two-step can keep A1 and replace A2 by the Hermite expansion of (19),

$$\mathbb{E}[\omega_j | u] = \rho_j \sigma_j u + h_{j2}(u^2 - 1) + h_{j3}(u^3 - 3u), \quad (27)$$

estimating the terms instead of testing them: each regime regression keeps the controls $\mathbb{E}[u^2 - 1 | D, Z]$ and, at order three, $\mathbb{E}[u^3 - 3u | D, Z]$ beside the Mills ratio, as in (19), and the stacked moments (24) carry them with their first-step correction. The gain becomes $\mathbb{E}[\omega_1 - \omega_0 | u] = \kappa u + \Delta h_2(u^2 - 1) + \Delta h_3(u^3 - 3u)$ with $\Delta h_k = h_{1k} - h_{0k}$, and $\kappa = \text{Cov}(\omega_1 - \omega_0, u)$ remains its linear part, the Hermite polynomials being orthogonal to u . The ATT and the ATU add the terms $\Delta h_k \mathbb{E}[H_k(u) | D, Z]$ to the expected gain of each unit, the ATE is $\mathbb{E}[X](\beta_1 - \beta_0)$ as before since the Hermite terms have mean zero, and the MTE line becomes the curve

$$\text{MTE}(p) = \mathbb{E}[X](\beta_1 - \beta_0) + \kappa v + \Delta h_2(v^2 - 1) + \Delta h_3(v^3 - 3v), \quad v = -\Phi^{-1}(p), \quad (28)$$

with its band from the covariance of all its coefficients. The influence functions are those of (26) with longer regime blocks.

The terms belong where they are needed. The ATT compares the mean outcome of the treated, which their regression fits whatever its regressors, with $\mathbb{E}[Y_0 \mid D = 1]$, which only the untreated regime predicts: it depends on the Hermite terms of the untreated regime and not on those of the treated one, and the ATU the other way round. Since `esrtest`, `normal` tests each regime, the order can be set by regime, `hermite(#1 #0)`: with the terms in the treated regime only, the ATT is exactly that of the line, and κ and the curve lose less precision. The order follows the cubic terms: three if they reject, two otherwise.

The price is precision. On the baseline design with $n = 2,000$ and 2,000 replications, where the line is right, the augmented estimates are unbiased with valid standard errors (ratios to the Monte Carlo standard deviation between 0.98 and 1.02), but their standard deviations are 2.4 to 3.4 times those of the line: 0.30 against 0.11 for the ATT, 0.46 against 0.14 for κ . On design C1, where the gain is $\kappa\{u + 0.5(u^3 - 3u)\}$ and the untreated regime is linear, the line keeps the ATT, ATU and ATE (biases below 0.02) but misses κ (0.78 for 1.08, coverage 0.65) and the MTE in the tails (coverage 0.35 at $p = 0.1$ and 0.67 at $p = 0.9$); the augmented two-step is unbiased, with a coverage between 0.92 and 0.95 (its standard errors fall three to six percent short on this design, a finite-sample effect), and with the terms in the treated regime only the standard deviation of the MTE at $p = 0.1$ falls from 0.65 to 0.38, below the root mean squared error of the line (0.61). At this sample size the line remains the better estimator of the average effects: the augmented route is for valid inference on κ and on the shape of the curve when A2 fails, read on the common support, where the polynomial does not extrapolate, with the semiparametric curve of Section 5.4 as its check. Choosing it after the Hermite test is a choice to state when reporting; `esrreport` proposes it when the test rejects and the instrument is strong.

5.4 Flexible control in P

The third tier keeps A1 and E, S, V, drops A2, and estimates the marginal treatment effect as the derivative of $\mathbb{E}[Y \mid X, P]$ in P on the support of the score, the local instrumental variable estimand of Heckman and Vytlacil (2005) that Brave and Walstrum (2014) and Andresen (2018) implement in Stata with local polynomials or splines in P , and that Cornelissen et al. (2016) review for the practitioner. The implementation in `esrmte`, `semipar` is the percentile-weights regression of Araar (2026c) on the probit score: at each percentile τ of P , a weighted regression of Y on X (centred), $X \times P$, P and P^2 with weights concentrated around the τ -quantile of P , and the MTE at that point read as $\hat{\beta}_P + 2p_\tau\hat{\beta}_{P^2}$. The bandwidth follows the derivative-aware rule of Araar (2026b), a pilot $h_0 \propto n^{-1/7}$ refined by an empirical mean-squared-error search on the target derivative, which widens the window where the curve is flat and narrows it where it bends. The curve is reported at the percentiles of the parametric line and next to it; the line is trusted where the curve follows it and the curve is trusted where they part, since it needs less. The effects on the common support can be integrated from the curve, and should agree with the restricted effects of `esrdiag` when the line is right. What this tier gives up is κ as one number and the extrapolation outside the support.

5.5 Other joint laws

A fourth tier keeps A1–A2 and replaces A3 by another joint law of (ω_j, u) : a copula between u and each regime error, with margins that can be skewed or heavy-tailed. It is not in the current command and is described here to fix its place. Its use is not to find the true law, which the data rarely pin down, but to report the effects under several laws and read the spread: when the ATU moves by less than its standard error across the Gaussian, Frank, Clayton and Gumbel copulas, the extrapolation into the tail of the untreated is not driving the result; when it moves by more, the restricted effect on the common support is the one to report. The band is the practicable counterpart of the bounds of Mogstad et al. (2018), computed on the same parametric skeleton, and it is the natural sequel to this paper.

6 Monte Carlo

The experiments of Section 4 share one baseline design, that of Araar (2026a) restated in Appendix B: $x, z, u \sim N(0, 1)$, $D = \mathbf{1}\{0.2 + 0.3x + 0.9z + u > 0\}$, $Y_1 = 1 + 1.2x + \omega_1$, $Y_0 = 0.5 + 0.4x + \omega_0$, with $\rho_1\sigma_1 = 0.78$ and $\rho_0\sigma_0 = -0.30$, hence $\kappa = 1.08$; $n = 5,000$ unless stated, 200 replications for the two-step-based tests and 100 where the likelihood is estimated at every replication. Each experiment switches one thing on. Table 2 collects the rejection rates at the 5 percent level and the quantities each section discussed; the scripts that produce every row are listed in Appendix B.

Three things stand out. The tests have their size: between 0.01 and 0.07 under the null across the blocks, the link test at order two and the Hausman contrast being conservative and the Hermite test slightly liberal at these sample sizes. Each test rejects what it is built to reject and not what its neighbour is built to reject: the link test is flat under skewed regime errors, the sufficiency test is flat under a nonlinear conditional mean, the Hermite test is flat under skewed errors, and the Hausman contrast is weak under a nonlinear conditional mean, where both routes move together. The one statistic that answers to two things is the γ contrast, which rejects both an omitted quadratic in the index and skewed regime errors; the link test beside it tells which, and the omitted quadratic costs both routes fifteen percent of κ while nothing in their own output shows it. The strength block shows the diagnostics tracking precision through the whole range, and the likelihood returning a confident wrong number at the end of it. The three single samples of Araar (2026b) behave in the same way (Section 4.5), and they are the validation cases of the command.

The standard errors of the estimators were checked on the same baseline design, the model holding exactly and every parameter strongly identified, with $n = 2,000$ and 5,000 replications (`tests/mc/mc_se.do`): for κ , the ATT, the ATU, the ATE and the MTE at three points by both routes, for $\hat{\kappa}_{DD}$, and for the slope of $\kappa(x)$ on design K, the ratio of the root mean square of the standard errors to the standard deviation of the estimates lies between 0.99 and 1.02 (precision ± 0.01), and the coverage of the 95 percent intervals between 0.947 and 0.956. The ATE alone sits at the top of that range, its standard errors one to two percent conservative.

7 Empirical illustration

The example is the one Stata’s manual uses to introduce `etregress` (StataCorp, 2025): the effect of union membership on the wages of young women in 1972, from the NLS extract `union3` (1,210 women with complete records, 253 union members) studied by Vella and Verbeek (1998). The outcome equation has `age`, `schooling` (`grade`), an SMSA indicator, `race` (`black`) and `tenure`; the selection equation has `south`, `black` and `tenure`, so that the only excluded instrument is the indicator of living in the South. It is a well-known, freely available and deliberately modest example, and this is why it is useful here: the manual fits the constrained model on the wage in levels, reports an ATE of 2.95 dollars an hour with $\rho = -0.57$ and a likelihood-ratio rejection of $\rho = 0$, and stops there. This section reads the same specification with the toolkit, in the order of Table 1. Tables 3 to 6 carry the numbers, all produced by `sec7_union3.do`; the text keeps to what they change.

What the tables say. Three things stand out, one from each block of tests. The selection equation (Table 4) is correctly specified and its instrument is strong, but it is a single indicator: the score takes few values, less than half of its variance is free of X , the Mills ratios have variance-inflation factors above two, index sufficiency cannot be tested and the only other candidate has no strength. Everything downstream rests on one exclusion restriction and on the normal form of the index, and the reader knows it before any effect is read. The joint law (Table 5) is where the manual’s number breaks: on the wage in levels, whose skewness is 2, the Hausman and γ

Table 2: Monte Carlo: size and power of the tests of Section 4, $n = 5,000$

Test	Design (what is switched on)	Rejection	Estimate	Comment
<i>Selection equation (§4.1), link test at orders 2 / 3 and γ contrast; 200 rep.</i>				
link	N (null)	0.01 / 0.05	1.08 / 1.09	size; conservative at order 2
link	Q: omitted quadratic in the index	1.00 / 1.00	0.92 / 0.91	$\hat{\kappa}$ two-step / FIML, true 1.08
link	C2 skewed errors	0.01 / 0.05	1.07 / 2.88	not confounded with A3
γ contrast	N (null)	0.07		size
γ contrast	Q: omitted quadratic	1.00		full Hausman 0.995
γ contrast	C2 skewed errors	1.00		A3 seen through γ
<i>Constant κ (§4.3), two-step Wald on $\theta_{1x} - \theta_{0x}$; 200 rep.</i>				
	K, slope $s = 0$ (null)	0.045	0.001	size
	K, $s = 0.1$	0.29	0.101	
	K, $s = 0.2$	0.72	0.201	
	K, $s = 0.4$	0.995	0.400	slope estimate, sd 0.08
	K, $s = 0.2$, $n = 2,000$	0.35	0.207	size 0.03 at $s = 0$
<i>Index sufficiency (§4.2), z_2 added, z_1 kept; 200 rep.</i>				
	S holds (null)	0.035	−0.000	size (0.045 at $n = 2,000$)
	Z: z_2 enters Y with −0.3	1.00	−0.300	direct effect recovered
	DH: double hurdle	1.00	0.237	rejects through the treated regime
	U3: A2 fails, S holds	0.07	0.012	not confounded with normality
<i>Strength (§4.4), γ_z weakened; 100 rep.; “estimate” = sd of $\hat{\kappa}$ two-step / FIML</i>				
	$\gamma_z = 0.9$: LR 1626, VIF 1.1	—	0.088 / 0.075	
	$\gamma_z = 0.45$: LR 542, VIF 1.4	—	0.14 / 0.12	
	$\gamma_z = 0.2$: LR 118, VIF 3.1	—	0.30 / 0.21	
	$\gamma_z = 0$: LR 1, VIF 96	—	1.9 / 0.63	FIML rmse 0.89: identification by form
<i>Normality by regime (§4.5); 100 rep. (200 for Hermite)</i>				
Hausman	N (null)	0.02	−0.006	κ contrast 0.07
Hausman	C2 skewed errors	1.00	−1.81	$\hat{\kappa}_{2S} - \hat{\kappa}_F$
Hausman	C1 nonlinear gain	0.52	−0.67	both routes wrong
Hermite	N (null)	0.07		0.045–0.075 over n , orders
Hermite	C2 skewed errors	0.07		shape untouched
Hermite	C1 nonlinear gain	0.94		through the treated regime

Rejection: share of replications rejecting at the 5 percent level. Estimate: the quantity named in the comment or, for the selection-equation block, the mean $\hat{\kappa}$ by route; for the constant- κ block, the mean slope $\hat{\theta}_{1x} - \hat{\theta}_{0x}$; for sufficiency the mean coefficient of z_2 in the treated regime; for normality the mean $\hat{\kappa}_{2S} - \hat{\kappa}_F$. Designs: Q, quadratic in the index as in §4.1; K, $\kappa(x) = 1.08 + sx$; S, DH, U3 as in §4.2; N, C1, C2 as in Araar (2026a).

Table 3: **union3**: the switching regression by route and by scale of the outcome

	log wage		wage in levels		manual
	FIML	two-step	FIML	two-step	etregress
$\hat{\rho}_1$	-0.583	-0.845	-0.364	-0.796	-0.575
$\hat{\rho}_0$	-0.386	-1.169	-0.904	-1.077	(one ρ)
$\hat{\sigma}_1, \hat{\sigma}_0$	0.312, 0.324	0.394, 0.465	2.191, 2.147	2.850, 2.444	2.019
$\hat{\kappa}$	-0.057 (0.071)	0.210 (0.152)	1.143 (0.412)	0.363 (0.931)	-
ATT	0.387 (0.062)	1.098 (0.249)	4.186 (0.194)	5.452 (1.390)	2.946 (0.275)
ATU	0.499 (0.107)	0.761 (0.195)	2.326 (0.695)	4.952 (1.323)	(= ATT)
ATE	0.475 (0.086)	0.831 (0.178)	2.715 (0.555)	5.057 (1.167)	2.946 (0.275)
ATT on the common support	0.388	1.099	4.193	5.460	

Standard errors in parentheses: observed information (FIML), stacked moments (two-step), and for the effects the influence function of the whole procedure (the estimation of the parameters, the averaging over the units and their covariance). The manual's column is the constrained model of **etregress** (common β up to the intercept, one σ and one ρ), for which ATT = ATE by construction. A two-step $\hat{\rho}_0$ below -1 is the implied ratio $\hat{\theta}_0/\hat{\sigma}_0$, not a constrained estimate.

Table 4: **union3**: the selection equation (Sections 4.1, 4.2 and 4.4)

Quantity	Value	Command
Excluded instrument: south , probit coefficient	-0.490 (0.093), Wald $\chi^2(1) = 27.5$	esrdiag
LR of the excluded instrument; pseudo- R^2 with / without	$\chi^2(1) = 25.7$; 0.046 / 0.025	esrdiag
$\text{Var}(P X) / \text{Var}(P)$	0.445	esrdiag
VIF of λ_1 on X (treated), of λ_0 (untreated)	2.12, 2.16	esrdiag
Range of $P(Z)$: treated, untreated; common support	[0.07, 0.76], [0.07, 0.70]; [0.07, 0.70]	esrdiag
Share extrapolated: treated, untreated	0.8%, 0.3%	esrdiag
Link test, order 2; order 3	$\chi^2(1) = 1.26$ ($p = 0.26$); $\chi^2(2) = 1.27$ ($p = 0.53$)	esrtest, spec
γ contrast, probit vs FIML: log wage; levels	$\chi^2(4) = 13.8$ ($p = 0.008$); 22.8 ($p < 0.001$)	esrtest, spec
Index sufficiency	not testable (one excluded instrument)	esrtest, suff
Second candidate instrument (msp): strength	Wald $\chi^2(1) = 0.7$	esrdiag

The probit does not depend on the scale of the outcome; the γ contrast does, through the FIML estimate.

Table 5: **union3**: selection on gains and the joint law (Sections 4.3, 4.5 and 4.6), log wage unless stated

Test	Statistic	Reading	Command
Pseudo-DiD, 2 strata: $C_1; C_0$; $\hat{\kappa}_{DD}$	0.084 (0.043); 0.098 (0.027); 0.44 (0.32)	$C_0 > 0$: $\rho_0\sigma_0 < 0$	esrtest, pdid
Pseudo-DiD, 4 strata: $C_1; C_0$; $\hat{\kappa}_{DD}$	0.084 (0.063); 0.205 (0.044); 0.78 (0.30)	same sign:	esrtest, pdid
$\kappa(x)$: slope on grade	0.028 (0.075), $p = 0.71$	common selection	nq(4)
Skewness of the outcome, members / non-members	0.4 / -0.4 (levels: 2.0 / 1.9)	no heterogeneity	esrtest, kappa
Hausman FIML vs two-step	$\chi^2(18) = 29.7$, $p = 0.040$ (levels: 55.8, $p < 0.001$)	A3 rejected	esrtest, normal
κ contrast, two-step - FIML	0.267 (0.130) (levels: -0.78 (0.90))		esrtest, normal
Hermite controls, joint; regime 1; regime 0; cubic terms	4.86 ($p = 0.30$); 4.01; 2.68; 3.43 ($p = 0.18$) (levels: 3.66, $p = 0.45$)	A2 not rejected	esrtest, normal
Link test (Table 4)	1.26 ($p = 0.26$)	index not rejected	esrtest, spec
Route to report	two-step, log scale		esrreport

Pseudo-DiD: $\hat{\kappa}_{DD} = C_1/\pi_1 + C_0/\pi_0$ with the X -adjusted Mills contrasts $\pi_1 = -0.25$ and $\pi_0 = 0.13$ (two strata), -0.40 and 0.21 (four strata), estimated with coefficients of variation of 20 to 29 percent: the command flags the ratios as far from normal and their standard errors as approximate.

contrasts reject trivariate normality with a clean index and clean Hermite controls—the pattern the Monte Carlo of Section 6 attributes to A3 alone—so that the likelihood is the estimator that moves and the ATE of 2.95 dollars rests on it; on the log wage the rejection weakens without

Table 6: `union3`, log wage: the effect along the score

<i>Expected effect by quintile of $P(Z)$ (<i>esrcurve</i>, <i>nq(5)</i>)</i>						
Quintile	1	2	3	4	5	
mean P	0.086	0.170	0.201	0.244	0.360	
two-step, members	1.22 (0.28)	1.16 (0.26)	1.14 (0.25)	1.07 (0.25)	1.02 (0.23)	
two-step, non-members	0.81 (0.19)	0.79 (0.20)	0.77 (0.20)	0.73 (0.20)	0.68 (0.19)	
two-step, all	0.85 (0.18)	0.85 (0.18)	0.84 (0.19)	0.81 (0.18)	0.80 (0.17)	
FIML, all	0.50 (0.09)	0.50 (0.09)	0.49 (0.09)	0.46 (0.09)	0.43 (0.08)	
<i>Marginal treatment effect (<i>esrmte</i>, <i>semipar</i>)</i>						
u	0.05	0.10	0.25	0.50	0.75	0.90
MTE line, two-step	1.18 (0.28)	1.10 (0.24)	0.97 (0.19)	0.83 (0.18)	0.69 (0.22)	0.56 (0.29)
MTE line, FIML	0.38 (0.07)	0.40 (0.06)	0.44 (0.06)	0.48 (0.09)	0.51 (0.13)	0.55 (0.17)
on the common support	no	yes	yes	yes	no	no
τ (quantile of the score); p_τ	0.05; 0.080	0.10; 0.089	0.25; 0.165	0.50; 0.193	0.75; 0.255	0.90; 0.316
PWR curve on the score	-3.40 (1.99)	-2.76 (1.54)	1.05 (0.97)	0.94 (0.23)	1.49 (0.33)	2.24 (0.54)

The score has 236 distinct values; the PWR bandwidths cover 0.05 to 0.21 of the range of P with effective samples of 300 to 700. The line is $m + \kappa \Phi^{-1}(1 - u)$ with $m = 0.831$ (two-step) and 0.475 (FIML).

disappearing ($p = 0.04$), and the route to report is the two-step. Selection on gains, finally, is not what these data show: $\hat{\kappa}$ is small and route-dependent, and the pseudo-DiD contrasts are both positive with only C_0 significant—across the strata of the index, the non-members most likely to join earn more than those least likely to. This is a common selection on the level of the wage, ρ_1 and ρ_0 of the same sign, which both routes confirm. The ratio $\hat{\kappa}_{DD}$, 0.44 with two strata and 0.78 with four, divides these contrasts by Mills contrasts known only to within a quarter of their size; the command says so, and the signs are what can be read.

What changes. The consequences are in Table 3. The two routes, which agree to the third decimal on the simulated samples of Section 6, differ here by a factor of nearly three on the log scale (ATT 0.39 against 1.10), and Table 5 says which route the tests support: the two-step, the published number being an ATE in dollars from a likelihood the data contradict. The two-step estimate is not a credible union premium for all that: 1.1 log points triples the wage, its implied $\hat{\rho}_0$ falls below -1 , and its standard error of 0.25 is the price of a single indicator instrument whose exclusion no test of the family can check here, and which a direct effect of living in the South on wages would violate. What the toolkit delivers on these data is the reason why this specification cannot pin the effect down, rather than a number. The heterogeneity along the score (Table 6, Figure 1) adds nothing on either route, which is what a κ near zero implies; the semiparametric curve adds a lesson instead. With 236 distinct values of the score between 0.08 and 0.38, the derivative of $\mathbb{E}[Y | X, P]$ in P is estimated from almost no variation: the percentile-weights regression returns 0.94 (0.23) at the median, close to the line, and swings to -3.4 and 3.5 at the ends. This is Assumption V failing in practice. With a single indicator instrument the third tier of Section 5.4 has nothing to work with, the parametric line is all there is, and the tests of the parametric assumptions are what protects the reader.

The reading. `esrreport` prints the four steps and concludes “route to report: two-step” with four notes: a single excluded instrument, sufficiency untestable and part of the identification by form; only one pseudo-DiD contrast significant, a common selection rather than a selection on gains; κ not different from zero, with the warning that a wide standard error is not evidence of $\kappa = 0$; the Hausman contrast rejecting with a clean index and clean Hermite controls, hence the two-step. None of this needed a new model; it needed the tests. A second application on a household survey with two instruments and a program with a clear selection on gains would show the toolkit at cruising speed; this one shows it doing what it is for.

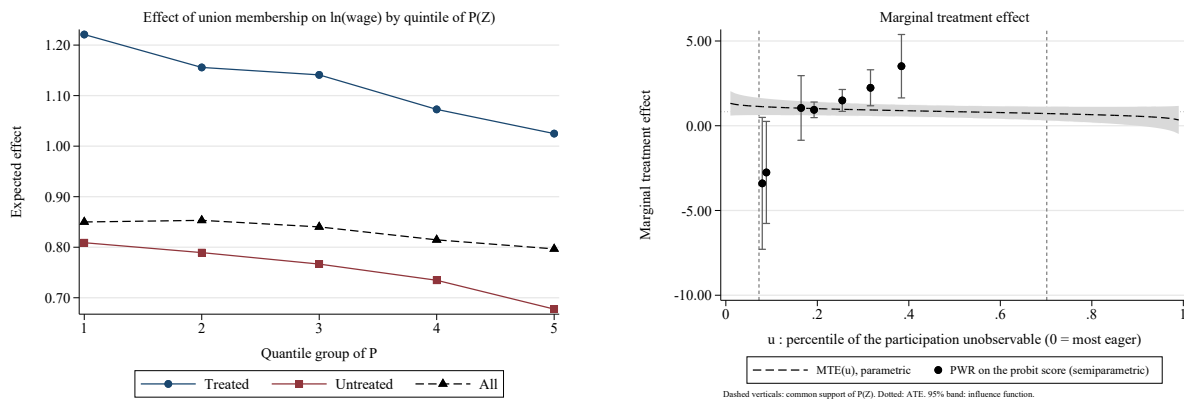


Figure 1: `union3`, log wage, two-step route. Left: expected effect of membership by quintile of the score $P(Z)$, members, non-members and all (`esrcurve`). Right: the parametric MTE line with its 95 percent band and the semiparametric curve on the probit score (`esrmte`, `semipar`); dashed verticals mark the common support, the dotted line the ATE. The curve is uninformative outside the middle of the score: a single indicator instrument leaves the derivative in P without variation to be estimated from (Assumption V).

8 The `esreg` command

8.1 What official Stata already does

The model of Section 2 is not new to Stata. Since version 14, `etregress` with the option `poutcomes` fits the “general potential-outcome model” (StataCorp, 2025): separate σ_j and ρ_j by treatment group, one outcome equation in which the treatment may be interacted with any covariate. When it is interacted with all of them, the likelihood is (22) and `esreg`, `method(fiml)` reproduces it to every printed digit on the samples of the manual (Section 7); `margins`, `predict(cte) subpop()` then gives the ATT, through the same formula $\mathbb{E}[x(\beta_1 - \beta_0) + (\rho_1\sigma_1 - \rho_0\sigma_0)\lambda_1 \mid D = 1]$ as the expected effect of the treated in Section 2.1, where κ appears without a name. The control-function estimator (`cfunction`) is the stacked-moment GMM of Section 5.2 extended to the variance parameters; the `twostep` option exists for the constrained model only. What the official command does not offer is everything this paper is about: κ as a parameter with a standard error, its heterogeneity in X and the heterogeneous regime laws, the two-step route for the general model with its exact variance, the tests of Section 4 (its only test is the Wald of $\rho_0 = \rho_1 = 0$), the diagnostics of the selection equation, the effects on the common support, the effect by quantile, and the MTE line and curve. `movestay` (Lokshin and Sajaia, 2004) fits the same likelihood with its own syntax and no post-estimation beyond `mspredict`; `msat` (Araar, 2026b) adds the effects, κ , the quantile profile and the MTE after it. Table 7 sets the three side by side.

8.2 The family

The design decision was to have one estimation command and a family of post-estimation commands, rather than one command that does everything: each tool of Sections 4 and 5 is a command with its own syntax, its own returned results and its own help, and the practitioner runs the ones the reading requires. Table 8 gives the syntax. `esreg` estimates the model by the route chosen in `method()`, computes the effects, κ and their standard errors, and stores the estimation automatically under the name `_esreg`, and under a second name if `store()` is given; every post-command works from `est(name)`, else from the current `e()` when it is an `esreg` estimation, else from `_esreg`, so that the post-commands can be run in any order after other

Table 7: The ESR in Stata: `etregress`, `movestay/msat` and `esreg`

	<code>etregress</code> <code>poutcomes</code>	<code>movestay</code>	<code>msat</code> (after <code>movestay</code>)	<code>esreg</code> family
Separate β_j , σ_j , ρ_j (FIML)	interactions	yes	–	yes
Two-step for the general model	– (CF GMM)	–	bootstrap	yes, stacked variance
Heterogeneous σ_j , ρ_j in covariates	–	–	–	<code>hetsigma()</code> , <code>hetrho()</code>
$\kappa = \rho_1\sigma_1 - \rho_0\sigma_0$ with s.e.	–	–	yes	yes
$\kappa(x)$ and its test	–	–	–	<code>kappa()</code> , <code>esrtest</code> , <code>kappa</code>
ATT, ATU, ATE with s.e.	<code>margins</code>	<code>mspredict</code>	yes	yes, plus common support
Effect by quantile of P	–	–	yes	<code>esrcurve</code>
MTE line / semiparametric curve	–	–	yes	<code>esrmte</code>
Strength, variation, support diagnostics	–	–	partly	<code>esrdiag</code>
Specification, sufficiency, normality, pseudo-DiD tests	–	–	–	<code>esrtest</code>
Factor variables, weights, <code>vce()</code>	yes	partly	weights	yes
<code>svy</code> : prefix; <code>svyset</code> weights, <code>hsize()</code>	ML only	–	<code>svyset</code> , <code>hsize()</code>	FIML; yes
Test of no selection on unobservables	Wald	LR	–	LR

Stata commands have overwritten `e()`, and several estimations can be kept and queried in one session. The post-commands protect the user's `e()`: they hold it, re-estimate what they need (the tests of Section 4.5, for instance, refit both routes), and restore it on exit. Factor variables and interactions are accepted everywhere; the `if/in` sample and the weights of the estimation are inherited by the post-commands through `e(sample)`, `e(wtype)` and `e(wexp)`.

Table 8: The `esreg` family: syntax

Command	Syntax and main options
estimation	<code>esreg depvar [indepvars] [if] [in] [weight], select(treatvar = varlist)</code> <code>[method(fiml twostep) hetsigma(varlist) hetrho(varlist) kappa(varlist)</code> <code>hermite(# [in]) vce(vctype) hsize(varname) nosvyset store(name) noeffects</code> <code>level(#) maximize options]</code> <code>svy: esreg ...</code> (FIML; see Section 8.3); replay: <code>esreg [, effects]</code>
predictions	<code>predict newvar [, statistic]: xb1 xb0 xbsel pr lambda1 lambda0 yc11 yc10 yc01</code> <code>yc00 effect kappa rhosig1 rhosig0 sigma1 sigma0 rho1 rho0; predict stub*,</code> <code>scores</code>
diagnostics	<code>esrdiag [, est(name)]:</code> strength (LR, per-instrument χ^2 , pseudo- R^2 increment), variation ($\text{Var}(P X) / \text{Var}(P)$, VIF of λ_j), support and extrapolated shares, effects on the common support; warnings
tests	<code>esrtest, spec [order(#) add(varlist)] (§4.1); esrtest, suff [keep(varname)]</code> <code>(§4.2); esrtest, kappa (§4.3); esrtest, normal (§4.5); esrtest, pdid [nq(#)</code> <code>reps(#)] (§4.6); all with [est(name) level(#)]</code>
effect by quantile	<code>esrcurve [if] [in], rank(varname) [nq(#) graph nose est(name) saving()]</code>
marginal treatment effect	<code>esrmte [if] [in] [, at(numlist) semipar graph est(name) saving()]</code>
reading	<code>esrreport [, est(name) nq(#) lrmin(#) vifmax(#) extrapmax(#) alpha(#)</code> <code>export(filename) notests]</code>

weight: `pweight` or `iweight` (fweights are refused, Section 8.3). `vce()`: `oim` (default), `robust`, `cluster` *clustvar*, `opg` for FIML; for the two-step, the stacked moments by observation (default), `cluster` *clustvar* or `svy`. `hetsigma()`, `hetrho()` are FIML options, `kappa()` and `hermite()` two-step options (Sections 2.2 and 5.3). The `bootstrap` and `jackknife` prefixes apply to `esreg` as to any estimation command.

Stored results. `esreg` returns `e(b)` and `e(V)` for the whole parameter vector (FIML: the seven equations y_1 , y_0 , the selection equation, $\ln \sigma_1$, $\ln \sigma_0$, $\text{atanh } \rho_1$, $\text{atanh } \rho_0$; two-step: the selection equation and the two regime blocks with their λ terms), and organizes what the post-commands

need: the matrix `e(effects)` (4×5 : ATT, ATU, ATE, κ by estimate, standard error, parameter and sampling variances and twice their covariance), the scalars `e(att)` ... `e(se_kappa)`, `e(sigma1)`, `e(rho1)`, ..., `e(supp_lo)`, `e(supp_hi)`, the coefficient blocks `e(b_sel)`, `e(b_1)`, `e(b_0)`, the support and mean Mills ratios, and the expanded variable lists `e(xvars)`, `e(zvars)`, `e(hetsigma)`, `e(hetrho)`, `e(kappavars)`. The post-commands are `rclass`: `esrtest` returns the statistics, degrees of freedom and p -values of its subtest and the tables it prints (`r(slopes)`, `r(coefs)`, `r(link)`, `r(gamma)`, `r(q)`, `r(hermite)`, `r(strata)`); `esrdiag` its diagnostics and the flags `r(weak)`, `r(form)`; `esrcurve` the table `r(table)`; `esrmte` the line `r(mte)` and the curve `r(mte_sp)`; `esrreport` the route `r(route)`, the numbered notes `r(note#)` and the figures they rest on.

The reading. `esrreport` runs the diagnostics and the tests on the stored estimation and prints the four steps of Section 7, with notes conditioned on the results: instrument strength and form (`lrmin()`, `vifmax()`), extrapolation (`extrapmax()`), the sign and significance of κ and the pattern of the pseudo-DiD contrasts, and the route to report, decided in the order of the assumptions—the index first (a rejected link test makes every downstream test unreadable), then sufficiency (a rejected exclusion biases both routes), then the conditional mean (Hermite controls: the augmented two-step when the instrument is strong, the semiparametric curve otherwise), then the joint law (Hausman: the two-step), else the likelihood. The notes are statements and questions, not verdicts; the thresholds are options, and each note names the test and the section it rests on. `export()` writes the figures and the notes to a plain-text sheet, the material an assistant, human or not, can read together with the context of the study; the rules stay in Stata, reproducible and auditable, and the interpretation stays outside.

8.3 Weights and survey design

Household surveys are the natural habitat of the model, and the family handles their weights at every step. `esreg` accepts `pweights` (the probit, the regime regressions and the likelihood are weighted, the FIML variance is then the sandwich, the two-step variance the weighted stacked moments, the effects weighted averages) and `iweights`; `fweights` are refused, since a unit of a survey stands for its sampling weight and is not a replicated record. When no weight is given and the data are `svyset`, the design weight is taken by default with a note (`nosvyset` declines it), and an explicit weight is used alone, never on top of the `svyset` one. The option `hsize(varname)` multiplies the weight by the household size, so that the effects are averages over individuals when the unit of observation is the household. The post-commands inherit the weights of the estimation.

The FIML route supports the `svy`: prefix: `esreg` declares the `svyb`, `svyj` and `svyr` properties, accepts the `iweights` the prefix passes, and `predict`, `scores` returns the seven equation-level scores of the likelihood, from which `svy` builds the linearized variance for the strata, clusters and finite-population corrections of the design—and the bootstrap, jackknife and BRR variances of its other `vce()` options. Since `svy` replaces `e(V)` after the command has returned, the effects, κ and their standard errors are recomputed from the linearized variance the first time the estimation is replayed, or on request with `esreg, effects; e(eff_vce)` says which variance they rest on. The two-step route does not take the prefix, for the same reason `etregress`, `twostep` does not: a sequential estimator has no single likelihood score for the prefix to linearize. It has its own design-based variance instead, `vce(svy)`: the influence functions of the whole procedure, aggregated through the strata, primary units, finite-population corrections and weights of `svyset`; `vce(cluster clustvar)` sums them within clusters. The effects follow the same aggregation by both routes, and after `svy`, `subpop()` the effects, the curves and the tests are those of the subpopulation. Under a design with weights only, the linearized standard errors of `svy: esreg` coincide with those of `esreg` with `pweights`, which is one of the checks of the validation file.

8.4 Validation and reproducibility

The specification of the commands is a Python reference library (`esreg/`, about a thousand lines: likelihood and analytic score, FIML, two-step with the stacked variance, effects, the tests and the diagnostics), written first; every number the Stata family prints was checked against it, value for value, on the simulated samples of this paper, on the samples of [Araar \(2026b\)](#), and on the manual’s `union3` and `drugexp`. The family was also checked against the official and community commands where they overlap: `etregress`, `poutcomes` with the treatment interacted with all covariates reproduces the FIML to the convergence tolerance and `margins`, `predict(cte)` the ATT; `etregress`, `cfunction` reproduces the two-step point estimates exactly and its robust variance to four digits, which validates Proposition 6 against an independent implementation of the same moment conditions; `movestay` and `msat` reproduce the likelihood and the effects. The Monte Carlo of the tests (Section 6) runs on the Python library.

The standard errors were checked against oracles that do not share the command’s formulas, in a do-file with tolerances and a pass-or-fail summary (`tests/test_effects_se.do`). The influence functions of the effects, of κ , of the slope of $\kappa(x)$, of the MTE line and of the augmented curve, by both routes and with the heterogeneous options, equal a brute force that re-runs the whole estimation with the weight of one observation moved (slope of one within 5×10^{-5} for the two-step and the convergence tolerance of `ml` for the likelihood); aggregated by cluster they give `ml’s vce(cluster)`, and through the design the `svy` prefix’s linearized variance; the pseudo-DiD equals an independent implementation in the reference library to 10^{-6} and the bootstrap of its whole procedure within a few percent on continuous indices. The Monte Carlo of the standard errors closing Section 6 and of the augmented two-step (Section 5.3) runs in Stata, from the do-files of the `tests/mc` folder.

The commands and the replication material of the paper share one repository, <https://github.com/aabdd12/esreg>. The package is installed from Stata with

```
. net install esreg, from("https://raw.githubusercontent.com/aabdd12/esreg/v1.0.0/")
replace
. net get esreg, from("https://raw.githubusercontent.com/aabdd12/esreg/v1.0.0/") replace
```

(the second line copies the data of the examples of `help esreg`; replacing `v1.0.0` by `main` installs the latest version): the ado files, the help files with their clickable examples and the dialog box; it will also be submitted to the SSC archive. The folder `replication` of the repository, which the installation does not copy (<https://github.com/aabdd12/esreg/tree/v1.0.0/replication>), holds the Python library and the Monte Carlo scripts with their outputs, the simulated samples with their generating scripts and seeds, the Stata do-files and logs that produce every table and figure, including Section 7, the raw results of the Monte Carlo of the standard errors and of the brute force, and the L^AT_EX source; the paths quoted in this paper are relative to it. A user who wants the command does not download the paper’s experiments, and a reader who wants to check a table finds it reproduced by one script, in Python and in Stata, from the same data. The package installed from the repository reproduces, on the `union3` sample, every number of Section 7 to the last printed digit.

9 Conclusion

The endogenous switching regression is fragile in the places where its selection equation is asked to do too much, and robust elsewhere; the purpose of this paper was to say where those places are and to give the practitioner a test for each. Ordered by what they cost, the assumptions are a correctly specified normal index, instruments that act only through it, enough variation in it, a linear conditional mean of the regime errors, and normality around that mean. The first three are properties of the selection equation and are tested on the probit and the two regime regressions; the fourth is tested by Hermite controls, which, kept in the estimation when it fails,

give the augmented two-step at a price in precision; the fifth by a Hausman contrast between the two routes, and it is the only one that can be dropped without loss, since the two-step route does not need it. The pseudo-DiD on the index closes the circle by reading selection on gains off a two-by-two table with the linear conditional mean alone, and the extension to $\kappa(x)$ turns the heterogeneity of that selection into a hypothesis.

Three lessons come out of the experiments and the example. The tests have their size and do not confound each other: the sufficiency test is flat under a nonlinear conditional mean, the Hermite test is flat under skewed errors, the link test is flat under both, and the contrast between the routes is the one statistic that answers to two things, with its neighbours saying which. The likelihood, when its joint law is wrong, returns a confident wrong number—the manual example’s ATE in dollars—and the two-step, which needs less, returns a less confident one, right if the exclusion of the instrument holds; the standard error of the two-step, computed exactly from the stacked moments, is the honest measure of what a single indicator instrument can bear, and on the manual’s data that exclusion is what no test can check. And the semiparametric tier, which needs the least, also needs the most from the data: without variation in the score it has nothing to work with, and on data like the manual’s the parametric line is all there is, which is why its assumptions must be tested rather than assumed.

Three things are left for later work. The copula tier of Section 5.5, which replaces trivariate normality by a family of joint laws and reports the untreated effect as a band across them, is described but not implemented; it is the practicable counterpart of the bounds of Mogstad et al. (2018) inside the parametric model. The exclusion of the instrument held out by the sufficiency test is the one assumption no test of the family checks (Section 4.2), and a sensitivity analysis to a direct effect of that instrument would give it a reading. And a second empirical application, on a household survey with two instruments and a program with a clear selection on gains, would show the toolkit where it was designed to work; the manual example shows it where it is needed. The commands, their Python reference implementation and the replication material of every table and figure are available with the paper (<https://github.com/aabbdd12/esreg>).

A Proofs

Propositions 1 and 3 are proved in the text. Throughout, $v = Z\gamma$, $\lambda_1(v) = \phi(v)/\Phi(v)$, $\lambda_0(v) = \phi(v)/\Phi(-v)$, and for $u \sim N(0, 1)$ the truncated moments are

$$\begin{aligned} \mathbb{E}[u \mid u > -v] &= \lambda_1(v), & \mathbb{E}[u^2 \mid u > -v] &= 1 - v\lambda_1(v), & \mathbb{E}[u^3 \mid u > -v] &= (v^2 + 2)\lambda_1(v), \\ \mathbb{E}[u \mid u < -v] &= -\lambda_0(v), & \mathbb{E}[u^2 \mid u < -v] &= 1 + v\lambda_0(v), & \mathbb{E}[u^3 \mid u < -v] &= -(v^2 + 2)\lambda_0(v), \end{aligned} \tag{29}$$

which follow from $\int_a^\infty u^k \phi(u) du$ by integration by parts ($\phi'(u) = -u\phi(u)$): $\int_a^\infty u\phi = \phi(a)$, $\int_a^\infty u^2\phi = a\phi(a) + 1 - \Phi(a)$, $\int_a^\infty u^3\phi = (a^2 + 2)\phi(a)$, evaluated at $a = -v$ and divided by $\Phi(v)$; the lower tail follows by symmetry.

A.1 Proposition 2 (identification by route)

(i) Under A1–A3, (Y, D) given Z has the density (22), the standard switching-regression likelihood of Maddala (1983, p. 224). Identification of γ comes from the marginal probit $P(D = 1 \mid Z) = \Phi(Z\gamma)$ under E and V (a column of Z outside X with non-zero coefficient rules out that $Z\gamma$ is a function of X alone, and the constant is identified by the normalization $\text{Var}(u) = 1$). Given γ , the conditional density of Y given $D = 1, X, Z$ is that of $X\beta_1 + \omega_1$ given $u > -Z\gamma$, a normal truncated through its correlation with u ; its mean $X\beta_1 + \rho_1\sigma_1\lambda_1(v)$ and variance $\sigma_1^2\{1 - \rho_1^2\lambda_1(v)(\lambda_1(v) + v)\}$ vary with v at fixed X by V, so that β_1 , $\rho_1\sigma_1$ and then σ_1^2 and ρ_1^2 are identified from the first two conditional moments, and the sign of ρ_1 from the first; the untreated regime is symmetric. The parameters identify $\kappa = \rho_1\sigma_1 - \rho_0\sigma_0$, the effects and the line (5) at every $p \in (0, 1)$, since the line is a function of the parameters alone.

(ii) Under A1–A2 only the conditional means are specified: $\mathbb{E}[Y \mid X, Z, D = 1] = X\beta_1 + \rho_1\sigma_1\lambda_1(v)$ and $\mathbb{E}[Y \mid X, Z, D = 0] = X\beta_0 - \rho_0\sigma_0\lambda_0(v)$, which is Proposition 1 with $W = 1$; γ is identified by the probit as in (i). The products $\rho_j\sigma_j$ are identified, hence κ , the effects and the line (5), which depend on the parameters only through $\beta_1 - \beta_0$, κ and γ . The conditional variance is not specified under A2, so σ_j and ρ_j are not separately identified; the residual variance of the regime regression identifies σ_j^2 only under the normal truncation formula, that is under A3. The line is a parametric extrapolation outside the support of $P(Z)$, where no conditional mean is observed: on the support it is the identified function $\mathbb{E}[Y \mid X, P, D = j]$ evaluated under A2, outside it is the assumption.

(iii) Under A1 alone, with v the single index, $P = \Phi(v)$ is identified and (X, P) carries the same information as (X, v) . By Heckman and Vytlacil (2005, Theorem 1), when $D = \mathbf{1}\{P > U\}$ with U uniform and independent of Z given X (here $U = \Phi(-u)$, and independence of u from Z given X is A1 with S), the marginal treatment effect $\text{MTE}(x, p) = \mathbb{E}[Y_1 - Y_0 \mid X = x, U = p]$ equals $\partial \mathbb{E}[Y \mid X = x, P = p] / \partial p$ at every p in the support of P given $X = x$, and $m_j(x, p) = \mathbb{E}[Y_j \mid X = x, P = p, D = j]$ are identified there as the observed conditional means. No functional form is used, so nothing is identified outside the support. If A2 holds in addition, $\mathbb{E}[Y_1 - Y_0 \mid X, U = p]$ is affine in $\Phi^{-1}(1 - p)$ with slope $-\kappa$ and κ is the common slope; if the conditional mean is not linear in u the slope varies with p and no single number is the slope. $\square$

A.2 Proposition 4 (separation of the two tests)

The Hermite terms. Let $\mathbb{E}[\omega_j \mid u] = \sum_{k \geq 1} a_{jk} H_k(u)$ with $H_1(u) = u$, $H_2(u) = u^2 - 1$, $H_3(u) = u^3 - 3u$ the probabilists' Hermite polynomials, and let S hold so that $\mathbb{E}[\omega_j \mid X, Z, u] = \mathbb{E}[\omega_j \mid u]$. Then $\mathbb{E}[\omega_1 \mid X, Z, D = 1] = \sum_k a_{1k} \mathbb{E}[H_k(u) \mid u > -v]$, and by (29), $\mathbb{E}[H_1 \mid u > -v] = \lambda_1$, $\mathbb{E}[H_2 \mid u > -v] = -v\lambda_1$, $\mathbb{E}[H_3 \mid u > -v] = (v^2 + 2)\lambda_1 - 3\lambda_1 = (v^2 - 1)\lambda_1$; below the threshold, $\mathbb{E}[H_1 \mid u < -v] = -\lambda_0$, $\mathbb{E}[H_2 \mid u < -v] = v\lambda_0$, $\mathbb{E}[H_3 \mid u < -v] = -(v^2 - 1)\lambda_0$. This is (19). Under A2, $a_{j2} = a_{j3} = 0$ whatever the law of ω_j given u beyond its conditional mean: a skewed ω_j with $\mathbb{E}[\omega_j \mid u]$ linear in u has zero Hermite coefficients, so the population regression of Y on X , λ_j and the two Hermite controls in regime j has zero coefficients on the controls, and the Wald statistic built on the stacked variance of Proposition 6 (which needs A1–A2 and finite fourth moments, not A3) is asymptotically $\chi^2(4)$. Under a failure of A2 with a_{j2} or a_{j3} non-zero, the population coefficients are non-zero and the statistic diverges; the regression on X and λ_j alone then omits a regressor correlated with λ_j , so its coefficient on λ_j converges to $\rho_j\sigma_j$ plus the omitted-variable term, and the likelihood, whose conditional mean is $X\beta_j \pm \rho_j\sigma_j\lambda_j$, is misspecified in the same direction: both routes are inconsistent for $\rho_j\sigma_j$ and κ , and neither line is the MTE.

The Hausman contrast. Let q be the common parameter and $\hat{q}_{2S}$, $\hat{q}_F$ the two estimators. Under A1–A3, both are regular $\sqrt{n}$ -consistent estimators with influence functions $\psi_{2S,i}$ and $\psi_{F,i}$ (the two-step from Proposition 6 mapped to q , the likelihood from the information equality, mapped to q by the delta method for $\rho_j\sigma_j = \tanh(c_j) \exp(a_j)$), so that $\sqrt{n}(\hat{q}_{2S} - \hat{q}_F) = n^{-1/2} \sum_i (\psi_{2S,i} - \psi_{F,i}) + o_p(1) \rightarrow N(0, \Omega_\Delta)$ with $\Omega_\Delta = \mathbb{E}[(\psi_{2S} - \psi_F)(\psi_{2S} - \psi_F)']$; the estimator (18) replaces ψ by its sample version evaluated at the estimates and is consistent for Ω_Δ by the law of large numbers and the continuity of ψ in the parameters. If Ω_Δ has rank r , the quadratic form with the Moore–Penrose inverse converges to $\chi^2(r)$, and $\hat{V}_\Delta^+$ with the estimated rank converges to it too. Since the likelihood is efficient under A1–A3, $\mathbb{E}[\psi_{2S}\psi_F'] = \mathbb{E}[\psi_F\psi_F']$ (the lemma of Hausman, 1978), and $\Omega_\Delta = V_{2S} - V_F$, the usual Hausman covariance; (18) is positive semi-definite in every sample while the difference of the two estimated variances need not be. Under a failure of A3 that keeps A1–A2, the two-step remains consistent for q (Proposition 1 with $W = 1$ uses the conditional mean only) while the likelihood converges to the pseudo-true value that minimizes the Kullback–Leibler distance to the normal density, which differs from q unless the true regime distributions are normal (the truncation formula that the likelihood applies to the observed regime is then wrong at every v , and the maximizer trades the fit of

the mean against the fit of the shape); the difference converges to a non-zero vector and the statistic diverges, while (18) remains a consistent estimator of the variance of the difference of two $\sqrt{n}$ -consistent estimators of different limits, so that the contrast is reported with a valid standard error. $\square$

A.3 Proposition 5 (pseudo-DiD)

First part. Under A1–A2, E and V, for a treated unit in stratum s of v , $\mathbb{E}[Y \mid X, v, D = 1] = X\beta_1 + \rho_1\sigma_1\lambda_1(v)$ by (29). In the within-treated regression of Y on X and the indicator T of the top stratum over the units of the top and bottom strata, $X\beta_1$ is fitted exactly by the regressors, so that the population coefficient of T is $\rho_1\sigma_1$ times the coefficient of T in the same regression of $\lambda_1(v)$, which is π_1 : $C_1 = \rho_1\sigma_1\pi_1$, with no approximation. The untreated regression gives $C_0 = -\rho_0\sigma_0\pi_0$ in the same way from $\mathbb{E}[Y \mid X, v, D = 0] = X\beta_0 - \rho_0\sigma_0\lambda_0(v)$. Since $\hat{\gamma}$ and the sample quantiles that cut the strata are consistent, the sample contrasts converge to their population values; π_1 and π_0 are non-zero by V and the strict monotonicity of λ_j as long as the adjustment for X does not undo the ordering of the strata, which the command shows by reporting them. Hence $C_1/\pi_1 \rightarrow \rho_1\sigma_1$, $C_0/\pi_0 \rightarrow -\rho_0\sigma_0$ and $\hat{\kappa}_{DD} \rightarrow \rho_1\sigma_1 - \rho_0\sigma_0 = \kappa$ for any fixed $G \geq 2$. The estimating equations of the whole procedure (probit score, quantile conditions of the cut points, the regressions) are just identified, and the influence function $-J^{-1}m_i$ is the M-estimation result used for Proposition 6, the indicators of the strata being smoothed for the derivatives in γ and in the cut points.

Second part. Under A2 alone with u continuous and any law F_u , $\mathbb{E}[\omega_1 \mid X, v, D = 1] = \rho_1\sigma_1 m_1(v)$ with $m_1(v) = \mathbb{E}[u \mid u > -v]$, and $\mathbb{E}[\omega_0 \mid X, v, D = 0] = \rho_0\sigma_0 m_0(v)$ with $m_0(v) = \mathbb{E}[u \mid u < -v]$. Both m_1 and m_0 are strictly decreasing in v : raising v lowers the threshold $-v$, which adds units with smaller u to the upper tail and removes units with the largest u of the lower tail, so that the conditional mean falls in both (formally, $\partial m_1/\partial v = -f_u(-v)\{m_1(v) + v\}/\{1 - F_u(-v)\} < 0$ since $m_1(v) > -v$, and symmetrically for m_0). Hence in the population $C_1 = \rho_1\sigma_1\pi_1^m$ and $C_0 = \rho_0\sigma_0\pi_0^m$, with π_j^m the X -adjusted contrasts of $m_j(v)$, negative when the adjustment for X preserves the ordering of the two strata; in the normal case $m_1 = \lambda_1$ and $m_0 = -\lambda_0$, so that $\pi_1^m = \pi_1$ and $\pi_0^m = -\pi_0$, and in general the sign of C_1 is that of $-\rho_1\sigma_1$ and the sign of C_0 that of $-\rho_0\sigma_0$. The signs need neither the normality of u (A1) nor the joint law (A3), only the monotonicity of a truncated mean. $\square$

A.4 Proposition 6 (variance of the two-step estimator)

Write $\vartheta = (\gamma', \beta'_1, \theta'_1, \beta'_0, \theta'_0)'$ and let $g_i(\vartheta)$ be the stacked moment (24). The two-step estimator solves $\sum_i g_i(\hat{\vartheta}) = 0$ exactly: the first block is the probit first-order condition, solved by $\hat{\gamma}$ alone; the second and third blocks are the normal equations of the regime regressions given $\hat{\gamma}$, solved by $(\hat{\beta}_j, \hat{\theta}_j)$. It is therefore the exactly identified GMM (M-) estimator with moment function g_i . Under A1–A2, E, S and V, $\mathbb{E}[g_i(\vartheta_0)] = 0$: the probit score has mean zero at the true γ under A1; the regime moments have mean zero because $\mathbb{E}[Y - r_{ji}(\beta'_j, \theta'_j)' \mid X, Z, D = j] = 0$ under A1–A2 with S (the conditional mean is exactly $r'_{ji}(\beta'_j, \theta'_j)'$ and r_{ji} is a function of (X, Z)). The Jacobian $G = \mathbb{E}[\partial g_i/\partial \vartheta']$ is block lower-triangular, with $\mathbb{E}[\partial s_i/\partial \gamma']$ (the negative probit information) in the first block, $-\mathbb{E}[D_i r_{1i} r'_{1i}]$ and $-\mathbb{E}[(1 - D_i) r_{0i} r'_{0i}]$ on the diagonal of the regime blocks, and the off-diagonal blocks $\mathbb{E}[D_i \partial \{r'_{1i}(y_i - r_{1i}(\beta'_1, \theta'_1)')\}/\partial \gamma']$, which carry $\partial \lambda_1(z_i \gamma)/\partial \gamma' = -\lambda_1(\lambda_1 + z_i \gamma) z_i$ and its mirror for λ_0 ; G is nonsingular by the identification of γ and the rank conditions of Proposition 1 (nonsingular $\mathbb{E}[r_{ji} r'_{ji} \mid D = j]$). With $\Omega = \mathbb{E}[g_i g_i']$ finite (finite fourth moments of the regime errors and of r_{ji}), the standard M-estimation expansion $0 = n^{-1/2} \sum_i g_i(\vartheta_0) + G\sqrt{n}(\hat{\vartheta} - \vartheta_0) + o_p(1)$ (Wooldridge, 2010, Theorem 12.3) gives $\sqrt{n}(\hat{\vartheta} - \vartheta_0) \rightarrow N(0, G^{-1}\Omega G^{-1})$, and (25) is consistent by the law of large numbers applied to $\hat{G}$ (the analytic or numerical derivative evaluated at $\hat{\vartheta}$) and to $\sum_i g_i g_i'/n$. Because the off-diagonal blocks of G are not zero, G^{-1} propagates the sampling error of $\hat{\gamma}$ into the regime blocks: this is the first-

step correction of Newey (1984), and setting these blocks to zero gives the regression-by-regime variance that ignores it. Because Ω is the outer product of the moments, the regime blocks of the variance are of the heteroskedasticity-robust form, which is what the truncation induces ($\text{Var}(Y \mid X, Z, D = j)$ depends on v under A3, and is unrestricted under A2). Finally $\kappa = \theta_1 - \theta_0$ and $\kappa(x) = W'(\theta_1 - \theta_0)$ are continuously differentiable functions of ϑ , so the delta method applies to $\hat{V}$. An effect is in addition an average over the units: expanding $\hat{\tau} = \sum_i w_i \tau_i(\hat{\vartheta}) / \sum_i w_i$ in $\hat{\vartheta}$ and in the empirical distribution gives $\hat{\tau} - \tau = \sum_i \phi_i + o_p(n^{-1/2})$ with ϕ_i of (26), whose first term is the parameter part and second the sampling part; both depend on (D_i, Z_i) , so that the variance $\sum_i \phi_i^2$ keeps their covariance. $\square$

B Monte Carlo design

Baseline. $x, z, u \sim N(0, 1)$ independent; $D = \mathbf{1}\{0.2 + 0.3x + 0.9z + u > 0\}$; $Y_1 = 1.0 + 1.2x + \omega_1$, $Y_0 = 0.5 + 0.4x + \omega_0$, $\omega_1 = 0.78u + \varepsilon_1$ with $\text{Var}(\varepsilon_1) = 1.3^2 - 0.78^2$, $\omega_0 = -0.30u + \varepsilon_0$ with $\text{Var}(\varepsilon_0) = 1 - 0.09$, so that $\sigma_1 = 1.3$, $\sigma_0 = 1$, $\rho_1 = 0.6$, $\rho_0 = -0.3$, $\kappa = 1.08$. The designs N, H, C1, C2 of Araar (2026a, Appendix B) are used as such (`mc_common.py`); H sets $\beta_1 = (2.5, 0.4)$ and $\omega_1 = \omega_0 = 0.5u + \varepsilon$; C1 keeps $\omega_0 = -0.30u + 0.95\varepsilon_0$ and sets $\omega_1 = \omega_0 + \kappa\{u + 0.5(u^3 - 3u)\} + 0.8\varepsilon_1$, so that the gain is cubic in u with linear part κ ; C2 draws ε_j from a centred log-normal with skewness parameter 0.5.

Switches of this paper. Q: $D = \mathbf{1}\{0.2 + 0.3x + 0.9z + 0.5(z^2 - 1) + u > 0\}$ with the baseline outcomes, the working probit on (x, z) (`mc_spec.py`). K: $\omega_1 = (0.78 + sx)u + \varepsilon_1$, so that $\kappa(x) = 1.08 + sx$ (`mc_kappa_x.py`). S, Z, DH, U3: two instruments, $D = \mathbf{1}\{0.2 + 0.3x + 0.7z_1 + 0.6z_2 + u > 0\}$; Z adds $-0.3z_2$ to both potential outcomes; DH replaces the selection by $D = \mathbf{1}\{0.3 + 0.3x + 0.9z_1 + e_1 > 0\} \cdot \mathbf{1}\{0.6 + 0.3x + 0.9z_2 + e_2 > 0\}$ with ω_j tied to e_1 ; U3 adds $0.6(u^3 - 3u)/\sqrt{6}$ to ω_1 (`mc_sufficiency.py`). Strength: $\gamma_z \in \{0.9, 0.45, 0.2, 0\}$ (`mc_strength.py`). Normality: N, C2, C1 (`mc_normal.py`). Standard errors (end of Section 6): N by both routes and K by the two-step, $n = 2,000$, in Stata (`tests/mc/mc_se.do`); the augmented two-step (Section 5.3): N and C1, $n = 2,000$, the line and the augmented routes on the same samples (`tests/mc/mc_aug.do`). Seeds are fixed in every script; the single-sample validation files `esr_link`, `esr_kx`, `esr_suff`, `esr_dh` are the draws with seed 2026 of Q, K ($s = 0.4$), S and DH. All scripts, the Python reference library and the Stata do-files that reproduce the tables are in the folder `replication` of the repository (Section 8.4).

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